9TH YEAR

#NATIONAL TECHNOLOGY INITIATIVE

UAV COMPETITIONS SPECIFICATION

2026

TEKNOFEST

AEROSPACE AND TECHNOLOGY FESTIVAL

VERSIONS

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CONTENTS

VERSIONS....1
CONTENTS....2
TABLES....5
FIGURES....6
DEFINITIONS....7
1. PURPOSE AND SCOPE OF THE COMPETITION....10
2. COMPETITION SCHEDULE....10
3. COMPETITION CATEGORIES....11
3.1. Fixed Wing Category....11
3.2. Rotary Wing Category....11
3.3. Free Mission Category....12
4. TERMS OF APPLICATION AND PROCEDURE....13
5. PROJECT PROCESS....16
5.1. Project Presentation Report....16
5.2. Preparation and Mission Video....17
6. FINANCIAL SUPPORTS....20
6.1. Project Support....20
6.2. Coorperation Supports....21
7. COMPETITION AREA....22
8. COMPETITION WEEK....22
8.1. General Process for All Categories....22
8.2. Matters Specific to Free Mission Category....24
8.3. Matters Specific to Fixed Wing and Rotary Wing Categories....26
8.3.1. General Matters....26
8.3.2. Payload Related Matters....26
8.3.3. Matters Related to Measurement of Distance to Target....27
8.3.4. Storage Box Related Matters....28
8.3.5. Matters Related to Time Measurement and Constraints....29
8.3.6. Autonomous Flight Related Matters....30
8.3.7. Matters Related to Scoring and Flight Invalidation....30
9. TECHNICAL RULES....32
9.1. Technical Controls....32

9.2. Flight Safety Control 32
9.3. UAV Specifications 34
9.3.1. Weight 34
9.3.2. Radio Control 34
9.3.3. Design 34
9.3.4. Emergency Power Cut-Off Mechanism 35
9.4. Rules to Follow During Flight 35
10. MISSIONS 36
10.1. Fixed Wing Category 36
10.1.1. International UAV Competition Fixed Wing Category Mission Descriptions 36
10.1.2. Inter High School UAV Competition Fixed Wing Category Mission Descriptions 42
10.2. Rotary Wing Category 47
10.2.1. International UAV Competition Rotary Wing Category Mission Descriptions ... 48
10.2.2. Inter High School UAV Competition Rotary Wing Category Mission Descriptions 54
10.3. International and Inter High School UAV Competitions Free Mission Category 59
11. SCORING 60
11.1. International UAV Competition Fixed Wing and Rotary Wing Categories 62
11.1.1. General Scoring Table 62
11.1.2. Mission Scoring Table 62
11.1.3. Scoring Parameters 62
11.2. Inter High School UAV Competition Fixed Wing and Rotary Wing Categories 64
11.2.1. General Scoring Table 64
11.2.2. Mission Scoring Table 64
11.2.3. Scoring Parameters 64
11.3. International and Inter High School UAV Competitions Free Mission Category 66
11.3.1. Evaluation Criteria and Evaluation Method 66
12. AWARDS 74
12.1. TUSAŞ Special Award 75
12.2. Payment of Awards 75
13. USE OF SUPPORT AND RETURNS 75
14. OTHER PROVISION 76
14.1. Administrative Provisions 76

14.2. Warnings Regarding Test Flights Before Competition Week 76
14.3. Security Measures 77
15. RELEVANT LEGISLATION 78
16. CONTACT 78

TABLES

Table 1: Competition Calendar.....10
Table 2: International UAV Competition Fixed Wing and Rotary Wing Categories General Scoring Table.....62
Table 3: International UAV Competition Fixed Wing and Rotary Wing Categories Mission Scoring Table.....62
Table 4: Inter High School UAV Competition Fixed Wing and Rotary Wing Categories General Scoring Table.....64
Table 5: Inter High School UAV Competition Fixed Wing and Rotary Wing Categories Mission Scoring Table.....64
Table 6: International and Inter High School UAV Competitions Free Mission Category Scoring Table.....66
Table 7: Free Mission Category Scoring Example.....73
Table 8: Awards 74

FIGURES

Figure 1: Screen Layout in Preparation and Mission Video 19
Figure 2: International UAV Competition Fixed Wing Category First Mission Flight Route.....38
Figure 3: International UAV Competition Second Mission Flight Route 41
Figure 4: Inter High School UAV Competition Fixed Wing Category First Mission Flight Route 44
Figure 5: Inter High School UAV Competition Second Mission Flight Route.....47
Figure 6: International UAV Competition Rotary Wing Category First Mission Flight Route.....50
Figure 7: International UAV Competition Second Mission Flight Route 53
Figure 8: Inter High School UAV Competition Rotary Wing Category First Mission Flight Route ..56
Figure 9: Inter High School UAV Competition Second Mission Flight Route.....59

DEFINITIONS

In this Specification, the following terms shall have the meanings set out below:

- **Advisor Teacher/Advisor Awards:** Awards given to the Advisor Teachers of Inter High School UAV Competition teams and Advisors (if any) of International UAV Competition teams who qualify for ranking awards.
- **Advisor Teacher:** A teacher who advises the team established by students of high school and equivalent school from all over the world and works in these schools and takes responsibility for the administrative and financial issues of the team and is called the Team Responsible in the competition.
- **Advisor:** An academician or a person possessing knowledge and experience in UAVs who advises teams participating in the International UAV Competition.
- **Advisory and Assessment Committee (AAC):** Committee assigned by TÜBİTAK to benefit from its opinions on the subjects regarding the UAV Competitions.
- **Announcement:** Announcement texts defining the competition subject, scope, application conditions, competition calendar, and special issues related to the competitions determined by TÜBİTAK and TEKNOFEST.
- **Autonomous Flight:** Flight managed by software capable of making its own flight decisions without requiring any UAV pilot intervention during the flight.
- **BİTO:** Directorate for Science and Society.
- **Department:** Department of Competitions and Organization within the Directorate for Science and Society, which conducts the UAV Competitions.
- **Ethical Violation:** Unethical situations such as copying and plagiarism (e.g., quoting from another team's report or video, or the report or video being the same or very similar).
- **Fail-safe System:** System designed to ensure a safe landing in any emergency during the flight, which is mandatory to be present on the control system.
- **Fixed Wing UAV:** Aircraft that takes all or a significant part of its lift from the non-moving wing system on it.
- **Flight Judge:** Authorized official responsible for supervising flight missions, evaluating mission execution, and making binding decisions regarding flight validity and safety during the competition.
- **Flight Zone:** Area where all flight missions take place and is limited to 120 meters vertically from the ground level and horizontally by the coordinates to be notified

before the competitions.

- **KYS:** Corporate Management System where all processes, including application, are managed.
- **Letter of Commitment:** Legal document signed by the Team Responsible and school/institution principals/directors of the high school students stating that all the responsibilities and rules of the competition have been accepted by all team members and submitted to TÜBİTAK officials during the registration in the competition area.
- **Letter of Consent:** Permission and rules acceptance document signed by parents/legal guardians of each team member under the age of 18 for the participation in the Competition and submitted to TÜBİTAK officials during the registration in the competition area.
- **Manual Flight:** Flight entirely under the control of the UAV pilot.
- **Mixed Team:** Team formed within the scope of the International UAV Competition by high school level students and associate, undergraduate, graduate, and doctoral students, where the Team Responsible is a university student and the number of high school students is less than half of the total team members.
- **Performance Awards:** Award given to the teams that participated in the competition without taking any financial support from TÜBİTAK and couldn't receive a ranking award but scored 55 points or more in the competition area.
- **Preparation and Mission Video:** Video recording with mission performance and preliminary information parts required for the selection of teams to be invited to the competition areas.
- **Project Presentation Report:** Report prepared according to the descriptions given in the "Guide for Project Presentation Report" with technical information regarding how to carry out the plans for design, production, mission requirements and information on the budget.
- **Ranking Awards:** Awards given to first, second, and third place winners with top scores for missions executed in the competition areas.
- **Regulation:** Regulation on Events to be carried out by the TÜBİTAK Directorate for Science and Society.
- **Rotary Wing UAV:** Aircraft such as helicopters and quadrotors which have more than one rotor, take all or a significant part of the carrying force from the rotor system and have the ability to rotate their wings.

- **Team Member:** Each person forming the team.
- **Team Responsible:** The person who takes the whole responsibility of the team and is teacher of the high school and equivalent school students from all over the world in the Inter High School UAV Competition teams and is university-level student in the International UAV Competition teams (including mixed teams),
- **Team:** A group consisting of the Team Responsible, members, and the Advisor (if any).
- **TEKNOFEST:** Aerospace and Technology Festival.
- **TÜBİTAK:** Scientific and Technological Research Council of Türkiye.
- **UAV Pilot:** Student controlling the UAV.
- **UAV:** Unmanned Aerial Vehicles desgined, produced and/or developed by competitors in accordance with the competition rules.
- **VTOL UAV:** Combined unmanned aerial vehicles capable of both vertical landing and take-off like helicopters and horizontal flight like fixed-wing aircraft.

1. PURPOSE AND SCOPE OF THE COMPETITION

International and Inter High School Unmanned Aerial Vehicle (UAV) Competitions organized by TÜBİTAK aim to direct students of high school and equivalent school and associate, undergraduate and graduate students from all over the world toward UAV technologies, develop their knowledge, skills and abilities by supporting their scientific and technical studies. The Competitions allow for flights with UAVs of all abilities and skills through different competition categories. Through different competition categories, flights based on all kinds of talents and skills with UAVs are encouraged. The International and Inter High School UAV Competitions also aim to create a meeting point in Türkiye for successful national and international students in UAV technology.

Competitions are organized by Department of Competitions and Organization in TÜBİTAK Directorate for Science and Society within the scope of TEKNOFEST. The creation, implementation, and supervision of technical rules are made by the Advisory and Assessment Committee (AAC) assigned by TÜBİTAK.

The Competition Specification covers the rules and obligations for the competitions to be held.

2. COMPETITION SCHEDULE

Table 1: Competition Calendar

Deadline for Application	March 15 2026
Submission of Project Presentation Reports	9 – 13 March 2026
Announcement of Project Presentation Report Results	6 – 10 April 2026
Deadline for Withdrawal from Competition and Team Member Changes	1 June 2026
Submission of Preparation and Mission Videos	8 – 12 June 2026
Announcement of Preparation and Mission Video Results	6 – 10 July 2026
Competition Week and Award Ceremony	August – September 2026
TEKNOFEST Award Ceremony	30 September – 4 October 2026

3. COMPETITION CATEGORIES

In 2026, the International and Inter High School UAV Competitions are held in three (3) different categories: Fixed Wing, Rotary Wing, and Free Mission.

3.1. Fixed Wing Category

- The UAVs of the teams competing in the Fixed Wing category perform the two missions specified below, the details of which are given under the heading "10.1. Fixed Wing Category".
- In the first mission, the autonomous flight capability of the fixed-wing UAV developed by the teams, the volume of the cylindrical storage box, the speed of removing the UAV from this box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.
- In the second mission, in addition to the requirements in the first mission, the UAV's payload carrying capacity and ability to make accurate drops on the target are evaluated. In the Inter High School UAV Competition, accurate payload release on targets with known locations is required. In the International UAV Competition, it is required to detect the location of the target through image processing and release the payload accurately on the targets.
- All UAVs competing in the International UAV Competition must carry two (2) payloads in the second mission. A payload painted red on the outside is released onto the 4 x 4 meter blue target, and a payload painted blue on the outside is released onto the 2 x 2 meter red target.
- All UAVs competing in the Inter High School UAV Competition must carry two (2) payloads in the second mission. A payload painted blue on the outside is released onto the target with a diameter of 10 m, and a payload painted red on the outside is released onto the target with a diameter of 2 m.

3.2. Rotary Wing Category

- The UAVs of the teams competing in the Rotary Wing category perform the two missions specified below, the details of which are given under the heading "10.2. Rotary Wing Category", and an optional third mission.
- In the first mission, the autonomous flight capability of the rotary-wing UAV developed by the teams, the volume of the rectangular or square prism-shaped

storage box, the speed of removing the UAV from this box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.

- In the second mission, in addition to the requirements in the first mission, the UAV's payload carrying capacity and ability to make accurate drops on the target are evaluated. In both the Inter High School UAV Competition and the International UAV Competition, it is required to detect the location of the target through image processing and release the payload accurately on the targets.
- In the International UAV Competition, in addition to the second mission, a third mission is defined following the release of payloads on the targets. In addition to the second mission, if teams go to the target where they dropped their first payload after dropping two payloads, pick up the payload on the ground, and drop it on the other target, they are deemed to have successfully completed the third mission. Performing the third mission isn't mandatory.
- All UAVs competing in the International UAV Competition must carry two (2) payloads in the second mission. A payload painted red on the outside is released onto the blue regular hexagonal target with each side measuring 2 meters, and a payload painted blue on the outside is released onto the red equilateral triangular target with each side length of 1 meter. Since the target placements in the field aren't made according to a specific order, teams perform the payload release process on the targets regardless of the order. The teams release the payload of the appropriate color on whichever target they detect first.
- All UAVs competing in the Inter High School UAV Competition must carry two (2) payloads in the second mission. A payload painted red on the outside is released onto the 2 x 2 meter blue target, and a payload painted blue on the outside is released onto the 1 x 1 meter red target. Since the target placements in the field aren't made according to a specific order, teams perform the payload release process on the targets regardless of the order. The teams release the payload of the appropriate color on whichever target they detect first.

3.3. Free Mission Category

- In the Free Mission category in the International and Inter High School UAV Competitions, the mission is determined by the teams within the safety and compliance limits defined in the specification. Details can be found under the

heading “10.3. International and Inter High School UAV Competitions Free Mission Category”.

- All types of UAVs (fixed wing, VTOL, flapping wing, hybrid, rotary wing, etc.) can compete in the Free Mission category. VTOL UAVs can’t compete in the Fixed Wing or Rotary Wing categories, they can only compete in the Free Mission category.

4. TERMS OF APPLICATION AND PROCEDURE

- Teams consisting of the students of high school and equivalent school from all over the world, high school students of BİLSEM, Youth Centers, Experimental Technology Workshops (DENEYAP), Science Centers and Science Workshops in Türkiye can apply to the **Inter High School UAV Competition** together with one (1) Team Responsible (Advisor Teacher).
- National and international associate, undergraduate and graduate students can apply to the **International UAV Competition** as a team together with one (1) Team Responsible (university level student) and Advisor, if any. Doctoral students may also apply within the scope of the International UAV Competition.
- National and international high school students and associate, undergraduate and graduate students can apply to the **International UAV Competition** by forming a **mixed team** with one (1) Team Responsible (university level student) and Advisor, if any. The number of high school students in the mixed teams must be less than half of the total number of members including the Advisor (for example, a maximum of 4 high school students in a team of 10 people).
- Teams consist of at least five (5) and at most ten (10) members, including the Team Responsible and the Advisor (if any). The number of members participating in the competition area and TEKNOFEST may be limited. In case of limitation, teams are informed.
- There must be no relationship accepted as a first-degree kinship or conflict of interest between the students in a team applying to the International or Inter High School UAV Competition and the Advisor Teacher or Advisor. In case of detection, the team is eliminated from the competition.
- A team member (student, Team Responsible, Advisor) can only take part in one (1) team across the International and Inter High School UAV Competitions, regardless of the category. In case of detection to the contrary, the teams in which the person is present are eliminated from the competitions.

- A team member can take part in only one (1) UAV Competition (International UAV Competition and Inter High School UAV Competition) organized by TÜBİTAK. In case of detection to the contrary, the teams in which the person is present are eliminated from both competitions.
- A team member applying to the International and Inter High School UAV Competition organized by TÜBİTAK may apply to other TÜBİTAK or TEKNOFEST competitions other than the said two (2) UAV Competitions.
- Any member of the team can't take a role as pilot for a different team in the International or Inter High School UAV Competition during the competition week.
- It is mandatory to have one (1) "Team Responsible" in all teams. The person called the Team Responsible must be an Advisor Teacher in high school teams, and a university student in university teams and mixed teams.
- The Team Responsible is responsible for carrying out all transactions of the team including application, report and video submission, taking responsibility for students in preparatory works and competition areas, and ensuring coordination.
- International UAV Competition teams may have one (1) "Advisor". The Advisor is an academician or an expert with knowledge and experience in UAVs who advises the team within the scope of the competition. If there is an Advisor, she/he needs to be registered during the application.
- It is mandatory for team members (excluding the Advisor Teacher and Advisor) to be students on the application date. In case of detection to the contrary, the team is eliminated from the competition.
- It is recommended that the team is formed from different classes and/or grade levels, taking into account the increase of students' scientific and technical gains, encouragement of coordinated teamwork, and the sustainability of the team in future years.
- Applications are made online via the application system at www.t3kys.com/en. No printed documents are requested from teams during the application.
- Team Responsible registers via the system on the application dates, adds the information of the team members and the Advisor (if any) correctly and completely to the system, and sends invitations to their e-mails.
- The invitation coming from the "My Team Information" section in the application system is accepted by the team members and the Advisor, and the registration is completed. Otherwise, the registration isn't completed.

- Team Responsible must assign herself/himself as the communication officer from the "Add Communication Officer" button while adding members.
- The responsibility regarding members who can't be reached due to incorrect or missing e-mail addresses or phone numbers belongs to the Team Responsible.
- The team name must be at most 20 characters long and unique to the team. It is recommended not to use generic names. TÜBİTAK may request a team name change from the team that applied later if there is more than one team applying with the same team names.
- Following the online application, teams upload the "Project Presentation Report" to the system (www.t3kys.com/en) within the dates specified in the competition calendar and enter other requested information into the system. Teams at or above the passing score which is determined by the AAC according to the Project Presentation Report results are selected eligible to continue the competition, and a "Preparation and Mission Video" is requested from these teams. Teams remaining below the passing score are eliminated from the competition.
- After the Project Presentation Report results are announced, teams that qualify to continue the competition by scoring at or above the passing score are subjected to budget evaluation. Financial project support is provided by TÜBİTAK to the teams determined as a result of the evaluation of the requested budgets. Teams that are at or above the passing score but aren't eligible to receive project support can upload their Preparation and Mission Videos using their UAVs made with their own means without receiving project support from TÜBİTAK and continue the competition.
- Teams that don't upload the Preparation and Mission Video are eliminated from the competition.
- The team member change including the Advisor (if any) must be performed before the June 1, 2026 through the "Discard a Member" option in the application system. The member change takes place if all the team members approve. The teams can't make members changes after the given deadline.
- Team Responsible must fill and sign the Letter of Commitment, stating that all the responsibilities and rules of the competition have been accepted, and submit the document to TÜBİTAK officials during the registration in the competition area. In additon, for each team member who is under the age of 18 on the July 10, 2026 (born on or after the July 11, 2008), the Letter of Consent filled and signed by the parents/legal guardians must be submitted during the registration in the competition

area, as well. The templates of the Letter of Commitment and the Letter of Consent are sent to Team Responsible via e-mail.

5. PROJECT PROCESS

5.1. Project Presentation Report

- It is the report where all requested information regarding team information, UAV information, and processes related to the mission to be performed within the scope of the applied category is presented in detail and uploaded to the system (www.t3kys.com/en).
- The Project Presentation Report is prepared via the "Project Presentation Report Template" published on the www.teknofest.org. The document named "Project Presentation Report Preparation Guide" published on the www.teknofest.org is used while preparing the Project Presentation Report.
- The report can be prepared in Turkish or English.
- The Project Presentation Report is evaluated by three (3) judges who are experts in their fields, and the final score of the team is obtained by taking the average of the scores given by the three (3) judges. Project Presentation Report evaluation criteria and score ranges are specified in the Project Presentation Report Preparation Guide.
- Reports exceeding seventeen (17) pages, including TEKNOFEST covers at the beginning and end of the template and annexes, aren't accepted by the system.
- In case it is detected that there is any element not specific to the team (plagiarism, copying, etc.) including mission information and design visuals of other teams, and in case of Artificial Intelligence (AI) usage in the reports sent by the teams, the report isn't evaluated and the team is eliminated from the competition. The team can't object to this situation.
- The list of teams that are at or above the passing score according to the Project Presentation Report evaluation results is announced on the www.teknofest.org.
- Team Responsibles of the teams whose score is at most twenty-five (25) points below the passing score can object to the evaluation results via the application system within three (3) calendar days from the announcement date. Teams with a difference of more than twenty-five (25) points between their score and the passing score can't object to the results. Objections submitted through channels other than the application system (such as e-mail) aren't accepted. Objections are evaluated by the

AAC and the results notified to the teams by e-mail. No second objection can be made to the objection evaluation results.

5.2. Preparation and Mission Video

- It is the video where the exact same mission to be performed in the competition area is performed and recorded in advance, along with the explanation of the design, project planning, and production processes of the UAV, and delivered by adding the video link to the system (www.t3kys.com/en) between the determined dates.
- Teams that achieve the minimum required score in the Project Presentation Report shall submit the Preparation and Mission Video.
- Teams prepare their videos in line with the "Preparation and Mission Video Preparation Guide" announced on the www.teknofest.org in the light of the specified purpose and rules, upload them to an online platform publicly, and add the video link to the application system (www.t3kys.com/en) on the specified dates.
- The evaluation is made over the video, no written report is requested.
- The Preparation and Mission Video can be a maximum of twenty (20) minutes. Parts of the video exceeding the specified duration aren't taken into evaluation.
- The following points are considered in the preparation of the Preparation and Mission Video:
  - **Flight Mission Section**

    Images belonging to the mission execution part are included, not exceeding ten (10) minutes in the Free Mission category and five (5) minutes in the Rotary Wing and Fixed Wing categories, with the voice-over of the team pilot or the responsible student in the team (excluding the Advisor and Advisor Teacher). It is expected that the exact same mission to be performed in the competition area (except for missions where realistic application is mandatory in the Free Mission category) including take-off and landing images, is performed without any interruption.

    No cutting or cropping can be made in the video in the Flight Mission and Execution section. In the Free Mission category, even if the mission is performed with more than one vehicle, no cutting or cropping should be made in the video. There should be natural and real images in the video.

    Teams that can't complete the Flight Mission and Execution section are deemed unsuccessful and eliminated from the competition regardless of their performance in the other section.

It may not be correct to expect a completely realistic mission for some missions in the Free Mission category. For example; missions such as kamikaze flights, expensive ammunition use are some of these missions. In this case, a realistic application is determined and these shots are made after obtaining approval from the AAC.

In Fixed Wing and Rotary Wing categories, for the flight of the UAV to be deemed successful, the aircraft must successfully complete the ∞ (infinite, figure-eight) maneuver, fly stably, and land without a crash in the video. If the maneuvers can't be fully completed, the flight takes place unstably, or an accident occurs during landing in the video, the team is deemed unsuccessful without scoring the remaining sections.

In Fixed Wing and Rotary Wing categories, Preparation/Briefing sections of teams aren't evaluated whose take-off, ∞ (infinite, figure-eight) maneuver movement, and landing movements aren't clearly seen in the flight section and exhibit an unsuccessful flight. These teams are eliminated from the competition with zero (0) points.

In the Free Mission category, if the UAV can't perform the flight or crashes during the flight, the mission is deemed unsuccessful and no scoring is made from other sections. The team is eliminated from the competition with zero (0) points.

In this section, the video needs to be divided into three (3) parts as "aircraft flight video", "simultaneous image of the pilot", and "ground station" as in Figure 1. Teams are responsible for shooting and showing the images simultaneously in the video. The pilot image must be a wide-angle shot where the team pilot performing the flight is clearly seen together with the stick movements, not just the stick movements. The images showing the ground station must be in the form of a screen recording taken directly from the ground station screen, not with an external camera. Videos of teams which aren't in the desired manner aren't evaluated.

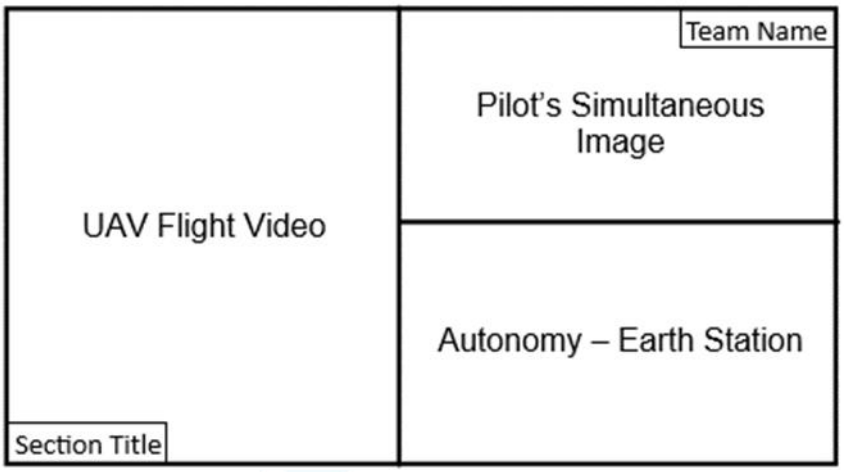

Figure 1: Screen Layout in Preparation and Mission Video

Preparation/Briefing Section

Students in the team, excluding the Advisor in International UAV Competition teams and the Team Responsible (Advisor Teacher) in Inter High School UAV Competition teams, provide information following the Flight Mission and Execution section, <u>not exceeding ten (10) minutes in the Free Mission category and fifteen (15) minutes in Fixed Wing and Rotary Wing categories</u>, and show time-lapse video clips. In this way, the evaluating judge has information about the conditions under which the team prepared its vehicle.

In this section; clips from the team's experience, infrastructure, opportunities they have, vehicle development stages, and preparations are showed.

In the Free Mission category, details regarding the mission and why this mission has been chosen are also mentioned.

- Teams competing in Fixed Wing and Rotary Wing categories can change the contents they wrote in the Project Presentation Report until the Preparation and Mission Video. It is expected to explain in the video why these changes have been made.
- Teams competing in the Free Mission category can make small-scale design changes which don't affect the mission scenario and the way the mission is performed specified in the Project Presentation Report, and don't cause the design to change completely. It is expected to explain in the video why these changes have been made.
- In accordance with the rules specified above, teams are required to explain the reasons for any changes made in their projects in the submitted video.
- The video must be shot in a high-light environment, in HD resolution, and in mp4 format. No music, including background music, should be used in the video.

- Video shootings are made in a safe area that doesn't create risk and danger. Team Responsible is responsible for the safety of the shootings and obtaining the necessary permissions.
- The video can be prepared in Turkish or English.
- The video is uploaded to an online platform and the link is added to the application system (www.t3kys.com/en) on the dates specified in the calendar. Teams that don't add the video link to the system are eliminated from the competition.
- The Preparation and Mission Video is evaluated according to the criteria determined by three (3) judges who are experts in their fields, and the final score of the team is obtained by taking the average of the scores given by the three (3) judges. Evaluation criteria and score ranges are specified in the Preparation and Mission Video Preparation Guide.
- Videos not prepared in accordance with the rules are considered unsuccessful and the team is eliminated from the competition.
- In case it is detected that there is any element not specific to the team (plagiarism, copying, etc.) in the videos of the teams and in case of Artificial Intelligence (AI) usage, the video isn't evaluated and the team is eliminated from the competition. The team can't object to this situation.
- The list of successful teams according to the evaluation is announced on the www.teknofest.org.
- Team Responsibles of the teams whose score is at most twenty-five (25) points below the passing score can object to the evaluation results via the application system within three (3) calendar days from the announcement date. Other teams can't object to the results. Objections transmitted through channels other than the application system (such as e-mail) aren't accepted. Objections are evaluated by the AAC and the results notified to the teams by e-mail. No second objection can be made to the objection results.

6. FINANCIAL SUPPORTS

6.1. Project Support

- It is the financial support requested by the teams in the Project Presentation Report stage and paid in cash to the Team Responsible by TÜBİTAK officials in the competition area in line with the evaluation of Advisory and Assessment Committee

provided that teams are successful in all stages, entitled to receive financial project support and invited to the competition area.

- Project support consists of material support given for goods, materials, and service procurement expenditures to be made to develop UAVs, and transportation support covering transportation costs to the city where the competition is held in case of being a finalist. The upper limit of project support (material + transportation) is 120,000 TRY.
- Teams prepare a budget table as specified in the Project Presentation Report Template for material support, provided that it doesn't exceed the support upper limit. The budget table is evaluated by TÜBİTAK and AAC, and if it is decided that the team receives support, the amount approved by TÜBİTAK in line with the evaluation of AAC is paid in cash to the Team Responsible in the competition area if the team become a finalist.
- The amount of transportation support to be provided to finalist teams is announced later. During the transportation support payment process, the amount of material support paid to the teams are checked and the transportation support is paid in a way that the total of transportation support and material support doesn't exceed 120,000 TRY.
- Teams that are at or above the passing score but aren't entitled to receive financial support aren't provided with a separate transportation support.
- In order for the project support to be paid, Team Responsible must fill and sign the Letter of Commitment, stating that all the responsibilities and rules of the competition have been accepted, and submit the document to TÜBİTAK officials during the registration in the competition area.
- For each team member who is under the age of 18 on the July 10, 2026 (born on or after the July 11, 2008), the Letter of Consent filled and signed by the parents / legal guardians must be submitted during the registration in the competition area, as well.
- The templates of the Letter of Commitment and the Letter of Consent are sent to Team Responsible via e-mail.

6.2. Coorperation Supports

- Teams can cooperate with the public and private sectors before and during the competition week to increase their opportunities and success.

7. COMPETITION AREA

- The location and date of the competition are announced later. The dimensions of the competition area and the runway length are also notified with the relevant announcement.
- The dimensions of the area to be scanned are planned to be approximately 30 x 100 meters. These dimensions may vary after the final competition area is determined. There is no buffer zone in the competition area.
- In Fixed and Rotary Wing categories, the distance between the two poles is planned to be 150 meters. However, this distance may vary according to the physical conditions of the competition area. The exact distance between the poles are announced separately after the competition area is determined.

8. COMPETITION WEEK

8.1. General Process for All Categories

The articles under this section are valid and binding for all teams competing in the Fixed Wing, Rotary Wing and Free Mission categories of International and Inter High School UAV Competitions.

- International teams must disassemble their UAVs to avoid being detained at customs in Türkiye.
- Teams must complete their registration in the competition area on the first day of the competition. Team Responsible is required to be present in the competition area during the registration process and throughout the entire competition period. Teams that fail to complete registration aren't allowed to compete in the flight area.
- Following the completion of final registration, teams must proceed to the technical control procedures. During this process, UAVs are examined by expert judges in terms of weight components, structural strength, mechanical mobility, and the safety of electronic equipment.
- Advisor Teachers of High School UAV Competition teams and Advisors (if any) of International UAV Competition teams aren't permitted to enter the technical control tent, interview tent, or flight area.
- UAVs equipped solely with cameras and authorized by TÜBİTAK and TEKNOFEST to perform flight operations for imaging purposes are entitled to conduct flights and capture footage at any time during the competition.

- In cases where it becomes difficult to continue the competition due to force majeure conditions such as heavy rain or strong winds, flight judges may suspend the competition. Once these conditions subside, the competition resumes. The threshold value for wind speed is set at 25 km/h.
- If flight judges perceive any safety risk during a UAV's flight, they may request the flight to be terminated. In such cases, the UAV must be switched to Fail-safe mode and perform a safe landing.
- In the image sensing process for the Fixed Wing and Rotary Wing categories, it is entirely the responsibility of the team to instantly present the labeled image of the detected target to the flight judges via the ground control station.
- The maximum flight altitude during the competition is 120 meters. However, for safety reasons and accurate observation, it is recommended that flights don't reach this altitude.
- It is the responsibility of the team to demonstrate to the flight judges that sufficient safety precautions have been taken regarding the materials, equipment, and flight techniques used during the competition. Compliance with the necessary safety measures are inspected.
- All flights are managed by flight judges and supporting officials appointed to assist flight judges. UAV flights are evaluated by the judges in accordance with the predetermined evaluation criteria.
- The pilot of the team must be one of the student members in the team. Advisor Teachers of Inter High School UAV Competition teams and Advisors (if any) of International UAV Competition teams aren't allowed to serve as team pilots. A pre-flight document is required from the team member who performs the piloting duty during the technical control stage. The name tags of the pilots whose pre-flight documents are found suitable are marked. The pilots whose tags aren't marked aren't permitted to enter the flight area.
- For safety reasons, TÜBİTAK may assign a Turkish Aeronautical Association (THK) pilot to be present in the competition area. The presence of THK pilots doesn't grant teams the right to receive piloting support.
- No verbal or physical intervention may be made toward TÜBİTAK and TEKNOFEST officials, flight judges, supporting staff, Turkish Aeronautical Association (THK) pilots, or other authorized personnel within the competition area. In case of such actions, the relevant team may be eliminated from the competition by TÜBİTAK.

8.2. Matters Specific to Free Mission Category

- Prior to the competition week, for reasons such as ensuring area safety, maintaining the competition schedule, and minimizing the preparation time of teams, grouping and flight order are determined by the Advisory and Assessment Committee (AAC), taking into account the missions to be performed by the teams. Teams can't object to the grouping and flight order.
- Each team is assigned a flight order and specific time slots in advance for the "Technical Control" and "Interview" procedures. Teams are required to complete all preparations before the competition.
- Teams that have made declarations under the categories of "Economy & Simplicity," "Innovation & Originality," and "Domestic Product Usage" are required to enter the Technical Control process with all declared components mounted on the UAV.
- Teams that successfully complete the Technical Control proceed to the interview tent on the first day of the competition according to the assigned flight order in order to receive scores for "Economy & Simplicity," "Innovation & Originality," and "Domestic Product Usage." Interviews are generally conducted in a question-and-answer format, and each team has fifteen (15) minutes. If deemed necessary, judges may revise the interview duration. During the interview, teams must explain the information previously declared under the "Economy & Simplicity," "Innovation & Originality" and "Domestic Product Usage" criteria.
- During the interview process, teams are obliged to support their declarations regarding the "Innovation & Originality" and "Domestic Product Usage" criteria with concrete and verifiable examples, including domestic contribution rates in software, mechanical, and electronic designs. If any declared original design, domestically produced part, or team-manufactured component isn't present on the UAV, no points are given for the relevant criterion. Teams that fail to declare or document this information when required aren't eligible to receive full points.
- Teams must have completed all necessary preparations in advance for the interview and scoring processes.
- <u>The procurement of all materials and equipment required for the execution of the declared mission is entirely the responsibility of the team</u>. Teams must determine their preparation times based on their mission requirements and comply with these schedules within the competition area.

- Flights in the Free Mission category are organized in a maximum of three (3) rounds, and teams are granted one flight right per round. The use of all three (3) flight rights isn't mandatory. For teams that use all three flight rights, only the flight with the highest score are taken into consideration for final evaluation.
- Teams are obliged to comply with the assigned flight order. Teams whose turn arrives but fail to complete their preparations within the designated time or fail to initiate their flight in the flight area are deemed to have used one flight right. Teams don't have the right to postpone their flights.
- Teams have to perform missions defined and declared in the report and video stages.
- In missions such as firefighting, military operations, search and rescue, or similar scenarios, a comparable or demonstration version of the mission may be performed if required. In such cases, prior approval must be obtained from the Advisory and Assessment Committee (AAC) before the competition.
- The flight must be initiated within five (5) minutes from the moment flight permission is granted by the flight judge. Teams that fail to complete necessary preparations, such as coordinate entry, in advance receive a warning for ineffective time management, and a point deduction are applied under the "Team Awareness" criterion. Teams that fail to initiate take-off within eight (8) minutes receive a second warning, and one flight right is deemed used. Teams that attempt take-off but fail may retry within their assigned eight (8) minute period.
- Except for cases caused by adverse weather conditions, teams whose UAVs can't fly in the competition area don't receive points from the evaluation criteria. In order to receive points, the UAV must successfully complete the take-off, flight, and landing phases. UAVs that fail to remain airborne or crash before achieving stable flight are deemed unsuccessful. Teams may not object to this decision. After resolving any technical issues or completing repairs, teams may wait for the next available flight order by re-entering the technical control process.
- Teams are strictly prohibited from conducting flights involving biological, chemical, or other hazardous substances that may endanger human health or safety within the competition area. Any flight that may pose a risk to human or environmental health, including large-scale explosions, is strictly forbidden. Instead, alternative materials and equipment such as paint systems or bead guns may be used.

8.3. Matters Specific to Fixed Wing and Rotary Wing Categories

8.3.1. General Matters

- VTOL UAVs can't apply under the Fixed Wing or Rotary Wing categories; they may only compete in the Free Mission category.
- It is mandatory that the design, sizing, and production processes of the unmanned aerial vehicles (UAVs) are carried out by the teams themselves. Teams participating in the competition with ready-to-fly UAVs purchased externally or UAVs with replica or copied designs are disqualified from the competition.
- The maximum loaded weight limit, including payloads, is 4 kg for both Fixed Wing and Rotary Wing UAVs.
- After the dimensional and weight inspections of the UAV storage box are completed prior to the flight, teams must place the UAV back into the storage box. In order to measure installation times, UAVs must be brought to the flight area unassembled and stored in the storage box. Propellers may be placed assembled inside the storage box, provided that sufficient space is available and safety conditions are met.
- Flight area dimensions are announced once the competition area is determined. The distance between two (2) reference poles in the flight area is determined in a manner that allows teams to safely perform their maneuvers and missions, and the planned distance between the two (2) poles is 150 meters.
- Teams are granted a total of four (4) flight rights for two (2) missions. It is not mandatory to use all flight rights. When calculating team scores, the flight with the highest score achieved for each mission is taken into consideration. Teams are required to comply with the assigned flight order.
- A maximum of four (4) team members from each team are allowed to enter the flight area. Advisor Teachers and Advisors aren't permitted to enter the flight area.
- There is no lower altitude limit for UAVs, provided that tje UAVs don't make contact with the ground. The upper altitude limit for UAV flights is based on the Above Ground Level (AGL) limit of 120 meters as determined by the Directorate General of Civil Aviation.

8.3.2. Payload Related Matters

- The content and dimensions of the payloads are at the discretion of the teams. All teams' UAVs are required to carry two (2) payloads in the second mission. The payloads must be identical in terms of weight and other characteristics, except for

color. The outer surface of each payload to be released onto different targets must be painted in the colors specified in Section “10. Missions”.

- Teams perform the first mission without payloads. Scoring for the first mission is based on the unloaded weight of the UAV. Within the scope of the second mission, UAVs are required to carry two (2) payloads and are allowed to make two (2) different payload attempts. Teams must determine the weights of the payloads in advance and declare them to the flight judges during the Technical Control process (e.g., first payload attempt 800 grams and second payload attempt 1200 grams).
- Teams that perform different payload attempts aren’t allowed to make any changes to the propulsion system, including but not limited to motors and propellers. Teams are only permitted to change batteries within the framework of the rules specified in Section “9.1 Technical Control”. Teams found to have used different motors, propellers, or electronic speed controllers (ESC) after technical control are disqualified from the competition.
- Payloads must be painted solely with the color specified in the Missions section. No additional elements such as markings, stripes, labels, or QR codes may be added to the payloads.
- Systems that don’t allow direct control of the payload, such as parachute systems, may be integrated into the payload. However, such systems must be the same color as the payload.
- No electronic components, whether performing active control functions or not, may be added to the payload.
- Payloads must be manufactured in a manner that prevents them from breaking apart upon impact with the ground. Distance measurements aren’t performed for payloads that lose their structural integrity, and even if they hit the target accurately, target hit scores aren’t taken into consideration.

8.3.3. Matters Related to Measurement of Distance to Target

- The distance between the payload and the target is measured based on the final resting position of the payload, not the point at which it initially falls. The distance to the target is measured from the center of the target.
- For Fixed Wing UAVs, target distance measurement is conducted up to twenty (20) meters from the center of the target, and for Rotary Wing UAVs, up to ten (10) meters from the center of the target. No distance measurement is performed for payloads

that fall outside these limits, and teams receive zero (0) points from this scoring criterion in the relevant mission.

- In the International UAV Competition, teams competing in the Fixed Wing and Rotary Wing categories use the same flight area. During image processing, rotary wing UAVs should not detect 4 x 4 meter and 2 x 2 meter targets positioned in the flight area for fixed wing UAVs. Similarly, fixed wing UAVs should not detect hexagonal and triangular targets positioned in the flight area for rotary wing UAVs when performing image processing.
- In the Fixed Wing and Rotary Wing categories of the International UAV Competition and the Rotary Wing category of the Inter High School UAV Competition, the positions of the targets within the flight area are randomly determined prior to each second mission flight. The targets are placed in suitable areas within the flight area and aren't positioned near the turning areas. Once the competition area location is announced, the boundaries of the area where teams must scan for image processing are announced too.

8.3.4. Storage Box Related Matters

- The storage box for Fixed Wing UAVs must be cylindrical. The cross-section of the cylinder must be circular and not elliptical. The dimensions and manufacturing method of the storage box are left to the discretion of the teams.
- The storage box for Rotary Wing UAVs must be rectangular or square prism-shaped. The dimensions and manufacturing method of the storage box are left to the discretion of the teams.
- The UAV storage box must have a robust structure and be of a quality that isn't damaged or deformed during transportation.
- The UAV storage box must be designed to be completely closed and must be of a size and weight that can be carried by one (1) person. All UAV components, including payloads, except for the ground control station and flight controller, must be placed inside the storage box. If any external component is detected to have been added to the UAV later and was not included in the storage box, a penalty score between zero (0) and five (5) points are applied to the team's total score.
- Equipment used for the assembly of the UAV such as pliers, screwdrivers, toolkits, and tape isn't required to be placed inside the storage box. No components may hang from or protrude outside the storage box. Add-ons such as handles and grips may be

used to facilitate transportation. Such add-ons are left to the discretion of the teams and aren't included in the calculated volume of the storage box.

- When calculating the volume of the storage box, the outermost dimensions are taken as the basis. Details such as the material and weight of the storage box are left to the discretion of the teams.

8.3.5. Matters Related to Time Measurement and Constraints

- In the International and Inter High School UAV Competitions, maximum time limits are defined for UAV installation, preparation for flight and mission flights.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time limit lose their flight right. The installation time starts when the judge says "start" and ends when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- The maximum flight time for the first mission is five (5) minutes, and the maximum flight time for the second mission is ten (10) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.

- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after completing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.

8.3.6. Autonomous Flight Related Matters

- There are score differences between fully autonomous, semi-autonomous, and manual operation of the UAV during take-off, flight, and landing phases. These differences are evaluated with the coefficient "c", which is the preliminary coefficient defined in the scoring section.

  c: Autonomous flight coefficient value is evaluated as follows:

  - 0.5 in case of manual take-off, manual flight and manual landing in the Inter High School UAV Competition,
  - 0.7 in case of manual take-off and/or manual landing, autonomous flight in the International and Inter High School UAV Competitions,
  - 1,0 in case of autonomous take-off, autonomous flight, and autonomous landing in the International and Inter High School UAV Competitions.

8.3.7. Matters Related to Scoring and Flight Invalidation

- The weight of the payloads to be carried by the UAVs is left to the discretion of the teams. UAVs with lower unloaded weight (mission 1), UAVs occupying less volume (mission 1), UAVs carrying heavier payloads (mission 2) and UAVs that accurately release the payload onto the target (mission 2) receive higher scores.
- Teams that fail to perform the required maneuver despite meeting other criteria such as autonomy or flight speed receive zero (0) points for the first mission. Teams that don't successfully complete the first mission aren't entitled to attempt the second mission.
- Although UAVs are required to carry payloads during the first mission, scoring for the first mission is based on the unloaded UAV weight.
- Teams that fail to return from the outside of the reference pole receive a –10 point penalty if the mission is completed. If a team fails to return from the outside of the reference pole for a second time, the flight is deemed unsuccessful.
- After landing, the structural and hardware integrity of the UAV must be preserved. As an exception, propeller breakage during landing is permitted and the flight is

considered successful. Teams whose UAVs are damaged or lose structural integrity may take a new flight queue number only after successfully passing technical control again.

- Flight judges are authorized to stop and cancel flights that significantly deviate from the designated flight pattern, navigate uncontrollably within the competition area, or pose a risk to the safety of the competition area and its surroundings.
- Teams are required to use the same motor and propeller used in the first mission while performing the second mission. UAVs are weighed before both missions. Except for mandatory situations such as accidents or crashes, no changes may be made to components affecting UAV's weight or the propulsion system.

9. TECHNICAL RULES

9.1. Technical Controls

- It is obligatory to comply with the decisions of TÜBİTAK and judges during the technical control process.
- All registered teams are required to undergo pre-flight technical control.
- Teams must participate in the competition using the UAV defined in the Project Presentation Report and the Preparation and Mission Video.
- All equipment that the UAV is required to carry during the flight must be brought to the technical control.
- The UAV must be brought to the technical control with its propellers removed, together with all of its components and connection mechanisms.
- The batteries to be used in the competition must be brought to the technical control together with spare batteries of the same capacity, model, and weight, if any. Batteries approved during the technical control may be replaced or charged within the competition area.
- Since UAVs that successfully pass the initial technical control may be damaged or crash during mission flights, technical controls continue throughout the competition days.
- A clearly defined “Radio Fail-Safe System (Fail-safe)” mode with specified technical characteristics is mandatory. Any justification to the contrary isn’t accepted during the technical control stage. UAVs that fail the radio fail-safe system control aren’t allowed to fly.
- Arm/Disarm operations of the motors must be performed using stick movements on the remote controller.

9.2. Flight Safety Control

- All UAVs must pass a pre-flight safety check. Teams whose UAVs fail the safety check aren’t allowed to fly.

In the flight safety control, the following points are especially checked:

- Compliance of the UAV with the technical drawings presented in the Project Presentation Report,
- Reliability of the UAV in terms of structural integrity, electrical and mechanical systems, and propulsion systems,

- All components being securely mounted on the UAV,
- All connections being made with safety wire, liquid adhesive and/or lock nuts in a quality and manner to prevent separation,
- Structural and connection integrity of the propeller, and compatibility of the motor with the rotation direction,
- Use of wires and connectors of sufficient thickness,
- The radio control range being sufficient for remote control operations such as UAV control and motor on/off,
- The radio control system having end-to-end encryption against interference from other teams' radio signals,
- All control mechanisms of the UAV being sufficient and not impairing flight safety,
- Radios being capable of automatically switching to fail-safe system mode in case of signal loss,
- Teams taking the necessary precautions since the teams are responsible for any problems that may arise while performing their flights in the competition area, as well as any problems that may arise from broadcast noise or pollution in the competition area.

Arm / Disarm operations of the motors must be performed with stick movements on the remote controller. The fail-safe system mode is tested on the ground by turning off the radio transmitter. When the fail-safe system is activated, the UAV receiver (if applicable) must select the following configuration:

In Fixed Wing UAV:

- Throttle cut,
- Full up altitude,
- Full right rudder,
- Full right aileron,
- Flaps full down.

In Rotary Wing UAV:

- Half throttle - controlled descent (return to launch (RTL) mode can't be selected),
- Throttle cut (in case the judge gives a command).

In VTOL UAV:

- The UAV switching to rotary wing mode,
- Half throttle - controlled descent,

- Throttle cut (in case the judge gives a command).

In the absence of any of the safety control systems specified above, the technical control judge must be convinced that the UAV can perform the required fail-safe function. Otherwise, the UAV isn't allowed to fly.

9.3. UAV Specifications

- Only fixed wing UAVs can compete in the Fixed Wing category of the competitions.
- Only rotary wing UAVs can compete in the Rotary Wing category of the competitions.
- All types of UAVs (fixed wing, VTOL, flapping wing, hybrid, rotary wing, etc.) can compete in the Free Mission category of the competitions. VTOL UAVs can't compete in the Fixed Wing or Rotary Wing categories and may only compete in the Free Mission category.

9.3.1. Weight

- In the Fixed Wing category of the competitions, the total take-off weight of the UAV can't exceed 4 kg.
- In the Rotary Wing category of the competitions, the total take-off weight of the UAV can't exceed 4 kg.
- In the Free Mission category of the competitions, the total take-off weight of the UAV can't exceed 25 kg.

9.3.2. Radio Control

- It is mandatory for the radio control system to have end-to-end encryption so that the UAV is minimally affected by environmental conditions.
- It is mandatory for the UAV to be capable of automatically switching to fail-safe system mode (within a maximum of five (5) seconds) in case of a signal loss experienced in the radio control unit.
- Any team performing any electromagnetic broadcast, jamming, spoofing, or frequency intervention that negatively affect the flight control, communication, GNSS reception, sensors, or mission performance of the competing UAV during the competition is diaqualified from the competition as soon as the situation is detected.

9.3.3. Design

- In Fixed Wing and Rotary Wing categories, it is mandatory for the design, sizing, and production processes of the UAVs to be carried out by the teams. Teams participating

in the competition with ready-to-fly UAVs they've purchased or UAVs with replica designs are disqualified from the competition.

- In the Free Mission category, the design of the UAV is not mandatory to be original. However, all ready-made parts and original parts must be clearly declared. Spare parts of the UAV can be brought to the competition area.

9.3.4. Emergency Power Cut-Off Mechanism

- An integrated Power On/Off switch that powers the system and ensures power cut-off without using an external circuit breaker must be present at an easily accessible point on the UAV to cut off the power of the UAV in an emergency.
- The switch must cut off the power within a maximum of two (2) seconds. Otherwise, the UAV isn't allowed to fly.
- During technical control, teams are asked to demonstrate that they can cut off the UAV's power within two (2) seconds. Powering the UAV without the approval of flight judges in the competition area is a cause for disqualification.
- The integration of the relevant power switch with the UAV must be explained in the Preparation and Mission Video.

9.4. Rules to Follow During Flight

- In the Free Mission category, the obligation to procure and bring the auxiliary platform and special equipment to be used for flight to the area belongs to the teams.
- In Fixed Wing and Rotary Wing Categories, the use of telemetry with end-to-end encryption is mandatory. UAVs with telemetry without end-to-end encryption have a risk of affecting other teams' UAVs. All responsibility that may be caused by telemetry belongs to the teams.
- Pre-tests of telemetry and image transmission systems must be performed before coming to the competition area.
- Based on any possible outcome, judges have the authority to terminate the flight due to safety reasons. In case the UAV is damaged/crashed before and/or during flights, detailed technical control is mandatory. The flight of the UAV that fails the technical control isn't allowed.
- No objection is accepted for any flight not allowed or canceled due to safety reasons.
- It is mandatory to wear helmets provided by TÜBİTAK during the flight.
- In missions requiring more than one pilot in the Free Mission category, a UAV pilot license is required for each pilot. Since the team pilot attempting to control different

UAVs simultaneously constitutes a risk in terms of flight safety, flight isn't allowed in such a case. Teams must assign a separate pilot for each UAV.

10. MISSIONS

10.1. Fixed Wing Category

- A mission definition has been determined in Fixed Wing categories in International and Inter High School UAV Competitions, and teams applying to these categories must perform the missions given below.
- All teams must carefully read "General Matters, Payload Related Matters, Matters Related to Measurement Distance to Target, Storage Box Related Matters, Matters Related to Time Measurement and Time Constraints, Autonomous Flight Related Matters, and Matters Related to Scoring and Flight Invalidation" under the heading "8.3. Matters Specific to Fixed Wing and Rotary Wing Categories" regarding the missions.
- Teams that successfully complete Mission 1 can proceed to Mission 2. Teams that can't successfully complete Mission 2 can't qualify to receive a Ranking Award.

10.1.1. International UAV Competition Fixed Wing Category Mission Descriptions

International UAV Competition Fixed Wing Category First Mission:

- In this mission, the autonomous flight capability of the fixed-wing UAV developed by the teams, the volume of the cylindrical storage box, the speed of removing the UAV from the cylindrical storage box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight, the unloaded weight of the UAV, and the time to complete the lap are parameters affecting the first mission score.
- Fixed-wing UAVs must perform the ∞ (figure-eight, infinite) movement by leaving the reference poles numbered 1 and 2 placed at the turning points in the flight area inside in every two (2) laps in an unloaded manner.
- UAVs fly the first mission unloaded. The first mission scoring is based on the unloaded UAV weight.
- Teams that can't perform the maneuver movement even though they meet other requirements (autonomy, fast flight, etc.) receive zero (0) points from the first

mission, and teams that don't successfully complete this mission don't have the right to try the second mission.

- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the first mission is five (5) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.

- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- UAVs take off from the take-off line. Following the take-off, the UAV proceeds directly to Pole 2 and performs a flight maneuver in the shape of infinity (∞) between two poles by turning from the outside of Pole 2. This maneuver is applied as two consecutive laps. After completing two laps, UAVs pass behind Pole 2 as specified in the figure, turn from the outside of the pole, and end their missions by crossing the start/finish line to complete the track.

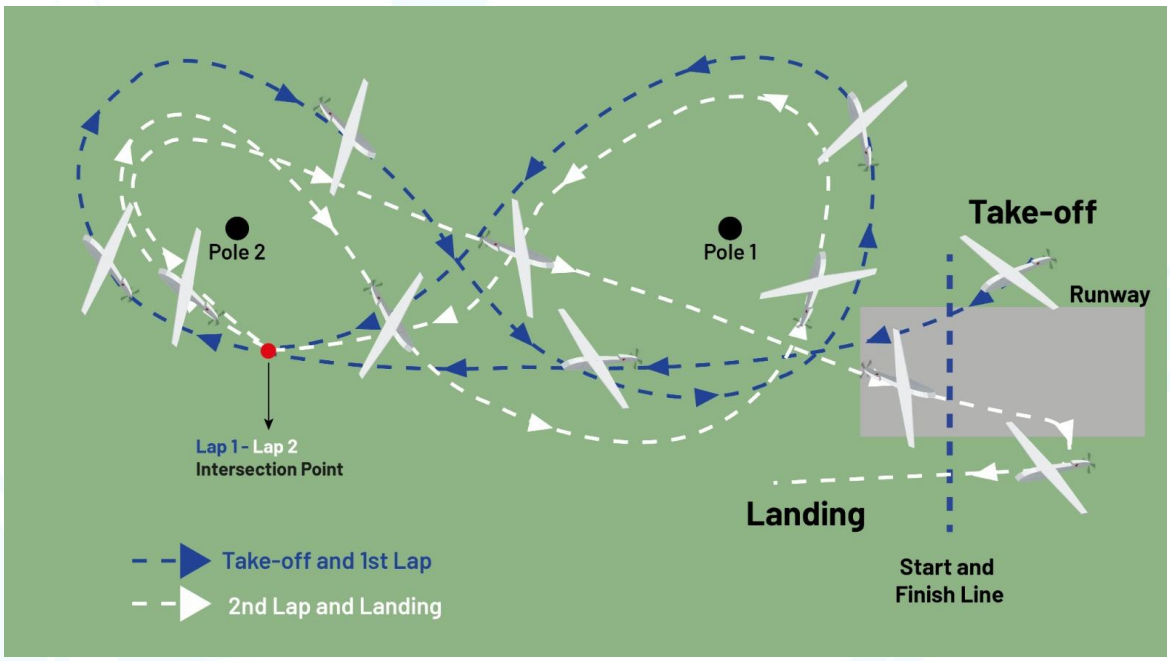

Figure 2: International UAV Competition Fixed Wing Category First Mission Flight Route

<u>International UAV Competition Fixed Wing Category Second Mission:</u>

- In this mission, teams are required to remove the fixed-wing UAV they developed from the cylindrical storage box and make it ready for flight, perform autonomous flight with the flight control software, detect the locations of two (2) different targets whose locations are unknown in the flight area via image processing technique, and drop two (2) different payloads, one (1) payload for each target.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight capability, the UAV's payload carrying capacity and unloaded weight, making an accurate drop as close as possible to the center of the target, and the time to complete the lap are parameters affecting the second mission score.

- UAVs are obliged to carry two (2) payloads within the scope of the second mission and can make two (2) different payload attempts in this mission. Teams are obliged to determine the weights of the said payloads in advance and notify the judges at the technical control (e.g., first payload attempt 800 grams and second payload attempt 1200 grams).
- A payload painted red on the outside is released onto the 4 x 4 meter blue target, and a payload painted blue on the outside is released onto the 2 x 2 meter red target.
- The distance of the payload to the target is measured based on where the payload last stops, not the place where it falls. The distance to the target is measured from the center of the target. For fixed-wing UAVs, target measurement is made up to 20 meters from the center of the target. No distance measurement is performed for payloads that fall outside these limits, and teams receive zero (0) points from this scoring criterion in the relevant mission.
- Image processing to be performed during the mission can be carried out from the air or ground without any lower altitude limitation. Teams are obliged to prove to the flight judges via the ground control station that they have detected the target via image processing method. Flight logs aren't requested from teams after the completion of flights. Even if a hit is achieved on the target without image processing, teams can't receive target hit points.
- Teams can use a single camera integrated into the auxiliary computer they use for image processing. Apart from this, using a second camera to view the area is strictly prohibited.
- Teams in Fixed Wing and Rotary Wing categories use the same runway. While performing image processing, fixed-wing UAVs should not detect the hexagonal and triangular targets positioned in the flight area for rotary-wing UAVs.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.

- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the second mission is ten (10) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- All teams must perform the target detection after turning from the outside of Pole 2 while performing their flights in the target detection phase in the second mission.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- Teams can make flight attempts with two (2) different payloads within the scope of the second mission. Teams trying different payloads must report all components and weight values they use in these flights at the technical control stage. Flights made with with payloads not declared in technical control are deemed invalid.
- Teams that perform different payload attempts aren't allowed to make any changes to the propulsion system, including but not limited to motors and propellers. Teams

are only permitted to change batteries within the framework of the rules specified in Section “9.1 Technical Control”.

- Teams are required to use the same motor and propeller used in the first mission while performing the second mission. UAVs are weighed before both missions. Except for mandatory situations such as accidents or crashes, no changes may be made to components affecting UAV’s weight or the propulsion system. Teams declare UAV components in technical control and state their flight weights. The highest weight of the UAV during the competition is included in the calculation.
- It is mandatory to obtain the approval of the flight judge before the flight regarding which direction the take-off is performed.

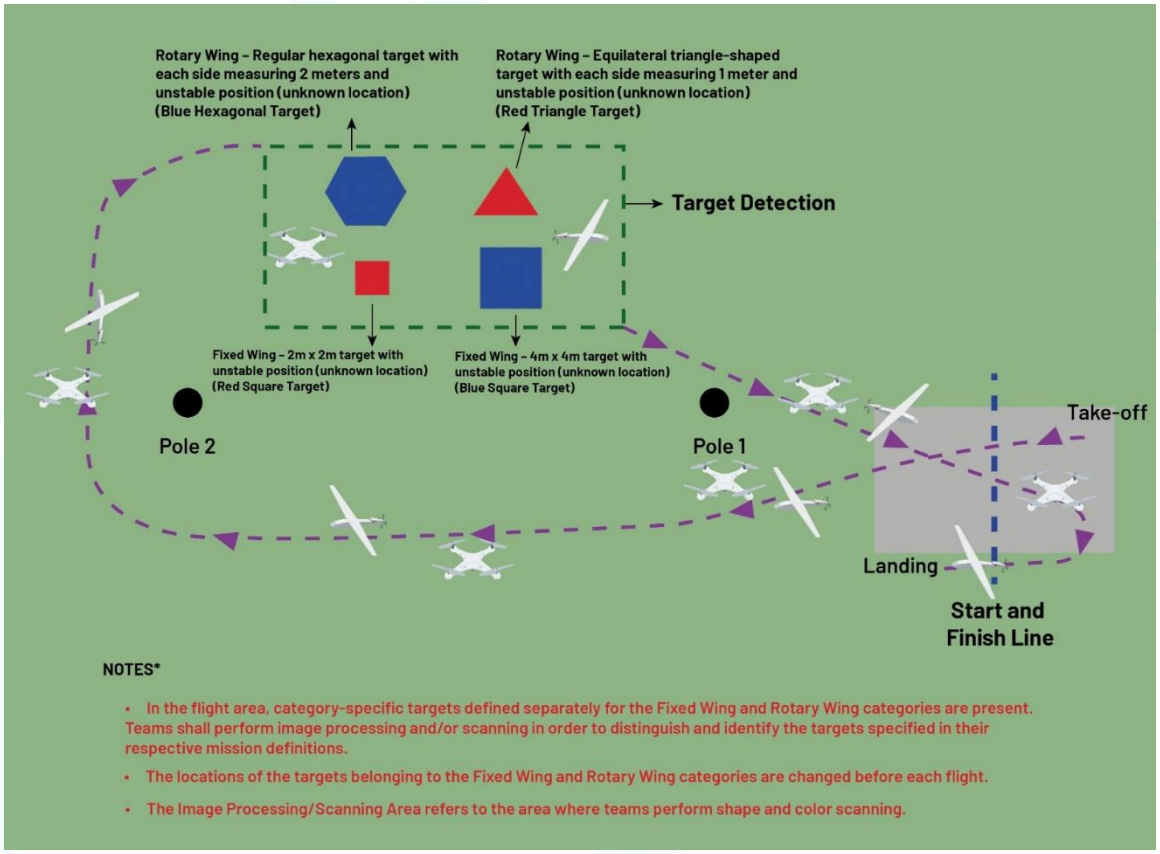

Figure 3: International UAV Competition Second Mission Flight Route

10.1.2. Inter High School UAV Competition Fixed Wing Category Mission Descriptions

Inter High School UAV Competition Fixed Wing Category First Mission:

- In this mission, the autonomous flight capability of the fixed-wing UAV developed by the teams, the volume of the cylindrical storage box, the speed of removing the UAV from the cylindrical storage box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight, the unloaded weight of the UAV, and the time to complete the lap are parameters affecting the first mission score.
- Fixed-wing UAVs must perform the ∞ (figure-eight, infinite) movement by leaving the reference poles numbered 1 and 2 placed at the turning points in the flight area inside in every two (2) laps in an unloaded manner.
- UAVs fly the first mission unloaded. The first mission scoring is based on the unloaded UAV weight.
- Teams that can't perform the maneuver movement even though they meet other requirements (autonomy, fast flight, etc.) receive zero (0) points from the first mission, and teams that don't successfully complete this mission don't have the right to try the second mission.
- In the Inter High School UAV Competition, the processes of obtaining coordinates and locations of targets to be used in missions requiring autonomy, entering them into the system, and retrieving coordinates from the system must be carried out by team members, excluding Advisor Teachers, during the preparation stage in the competition area.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.

- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the first mission is five (5) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- UAVs take off from the take-off line. Following the take-off, the UAV proceeds directly to Pole 2 and performs a flight maneuver in the shape of infinity (∞) between two poles by turning from the outside of Pole 2. This maneuver is applied as two consecutive laps. After completing two laps, UAVs pass behind Pole 2 as specified in the figure, turn from the outside of the pole, and end their missions by crossing the start/finish line to complete the track.

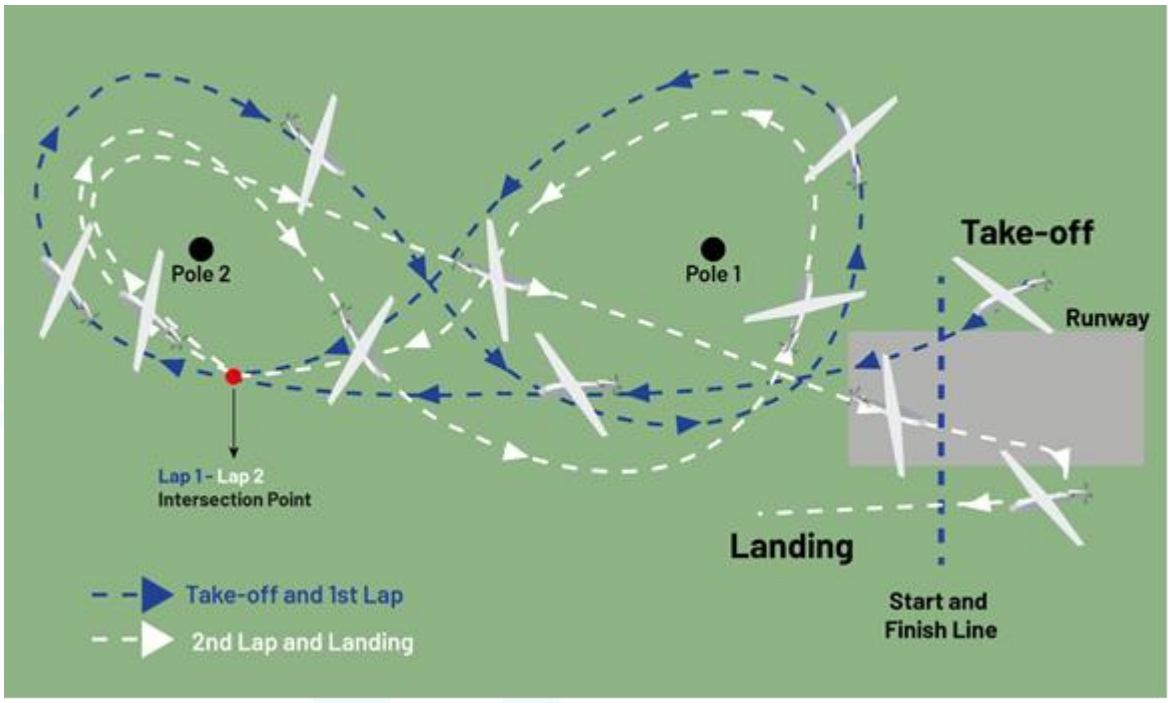

Figure 4: Inter High School UAV Competition Fixed Wing Category First Mission Flight Route

<u>Inter High School UAV Competition Fixed Wing Category Second Mission:</u>

- In this mission, teams are required to remove the fixed-wing UAV they developed from the cylindrical storage box and make it ready for flight, perform autonomous flight with the flight control software, and drop two (2) different payloads, one (1) payload for each target, onto targets with diameters of 10 meters and 2 meters whose locations are known in advance in the flight area.
- The autonomous flight capability of the UAV, the volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, the UAV's payload carrying capacity and unloaded weight, making an accurate drop as close as possible to the center of the target, and the time to complete the lap are parameters affecting the second mission score.
- UAVs are obliged to carry two (2) payloads within the scope of the second mission and can make two (2) different payload attempts in this mission. Teams are obliged to determine the weights of the said payloads in advance and notify the judges at the technical control (e.g., first payload attempt 800 grams and second payload attempt 1200 grams).
- Since the teams are provided with the locations of the targets, the color of the targets isn't specified.

- The distance of the payload to the target is measured based on where the payload last stops, not the place where it falls. The distance to the target is measured from the center of the target. For fixed-wing UAVs, target measurement is made up to 20 meters from the center of the target. No distance measurement is performed for payloads that fall outside these limits, and teams receive zero (0) points from this scoring criterion in the relevant mission.
- Teams in Fixed Wing and Rotary Wing categories use the same runway.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.

- The maximum flight time for the second mission is ten (10) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- All teams must perform the target detection after turning from the outside of Pole 2 while performing their flights in the target detection phase in the second mission.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- Teams can make flight attempts with two (2) different payloads within the scope of the second mission. Teams trying different payloads must report all components and weight values they use in these flights at the technical control stage. Flights made with with payloads not declared in technical control are deemed invalid.
- Teams that perform different payload attempts aren't allowed to make any changes to the propulsion system, including but not limited to motors and propellers. Teams are only permitted to change batteries within the framework of the rules specified in Section "9.1 Technical Control".
- Teams are required to use the same motor and propeller used in the first mission while performing the second mission. UAVs are weighed before both missions. Except for mandatory situations such as accidents or crashes, no changes may be made to components affecting UAV's weight or the propulsion system. Teams declare UAV components in technical control and state their flight weights. The highest weight of the UAV during the competition is included in the calculation.

- It is mandatory to obtain the approval of the flight judge before the flight regarding which direction the take-off is performed.

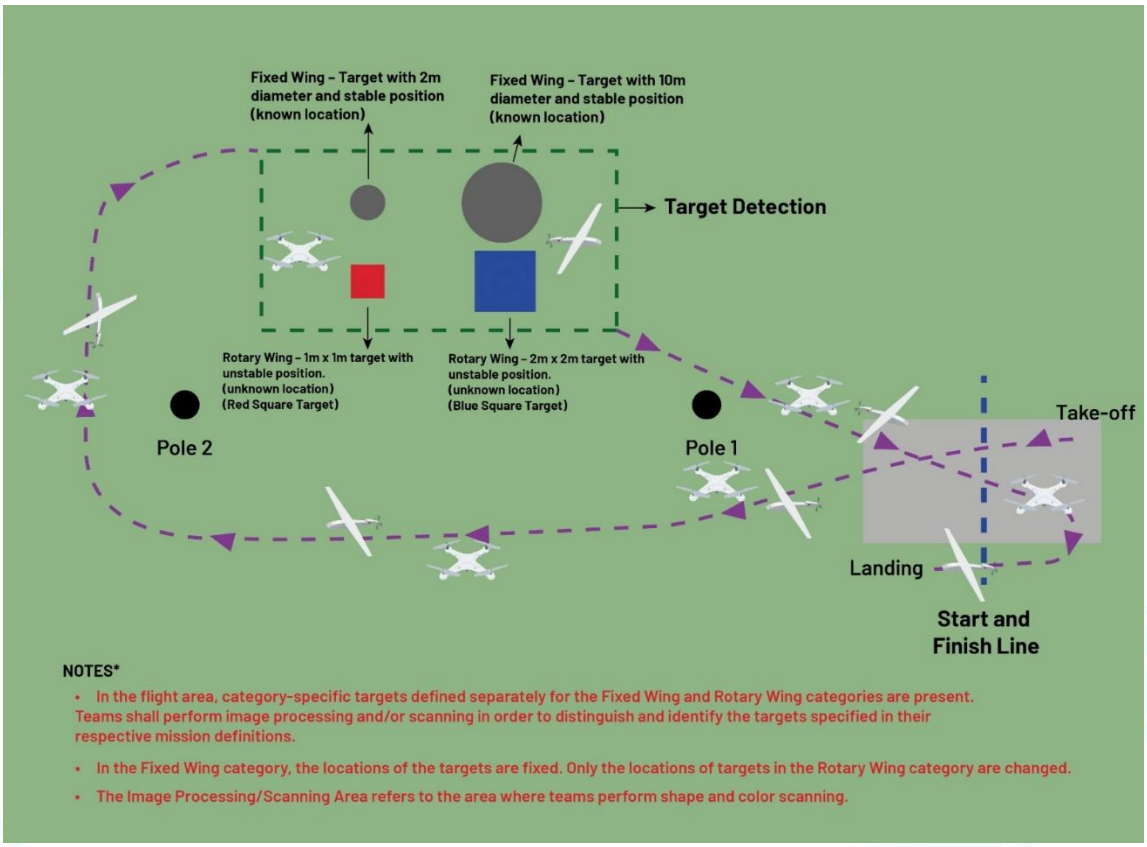

Figure 5: Inter High School UAV Competition Second Mission Flight Route

10.2. Rotary Wing Category

- A mission definition has been determined in the Rotary Wing categories in International and Inter High School UAV Competitions, and teams applying to these categories must perform the missions given below.
- All teams must carefully read “General Matters, Payload Related Matters, Matters Related to Measurement Distance to Target, Storage Box Related Matters, Matters Related to Time Measurement and Time Constraints, Autonomous Flight Related Matters, and Matters Related to Scoring and Flight Invalidation” under the heading "8.3. Matters Specific to Fixed Wing and Rotary Wing Categories” regarding the missions.
- Teams that successfully complete Mission 1 can proceed to Mission 2. Teams that can’t successfully complete Mission 2 can’t qualify to receive a Ranking Award.

10.2.1. International UAV Competition Rotary Wing Category Mission Descriptions

International UAV Competition Rotary Wing Category First Mission:

- In this mission, the autonomous flight capability of the rotary-wing UAV developed by the teams, the volume of the rectangular or square prism-shaped storage box, the speed of removing the UAV from the rectangular or square prism-shaped storage box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight, the unloaded weight of the UAV, and the time to complete the lap are parameters affecting the first mission score.
- Rotary-wing UAVs must perform the ∞ (figure-eight, infinite) movement by leaving the reference poles numbered 1 and 2 placed at the turning points in the flight area inside in every two (2) laps in an unloaded manner.
- UAVs fly the first mission unloaded. The first mission scoring is based on the unloaded UAV weight.
- Teams that can't perform the maneuver movement even though they meet other requirements (autonomy, fast flight, etc.) receive zero (0) points from the first mission, and teams that don't successfully complete this mission don't have the right to try the second mission.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.

- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the first mission is five (5) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- UAVs take off from the take-off line. Following the take-off, the UAV proceeds directly to Pole 2 and performs a flight maneuver in the shape of infinity (∞) between two poles by turning from the outside of Pole 2. This maneuver is applied as two consecutive laps. After completing two laps, teams pass behind Pole 2 as specified in the figure, turn from the outside of the pole, and end their missions by crossing the start/finish line to complete the track.

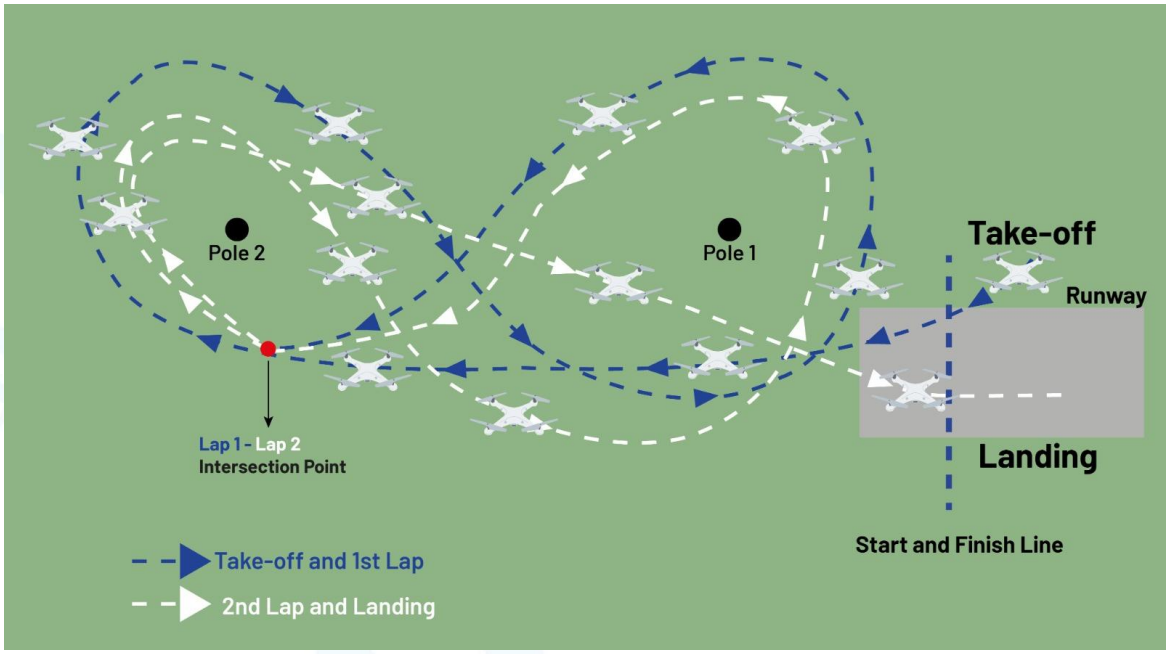

Figure 6: International UAV Competition Rotary Wing Category First Mission Flight Route

<u>International UAV Competition Rotary Wing Category Second Mission:</u>

- In this mission, teams are required to remove the rotary-wing UAV they developed from the rectangular or square prism-shaped storage box and make it ready for flight, perform autonomous flight with the flight control software, detect the locations of two (2) different targets whose locations are unknown in the flight area via image processing technique, and drop two (2) different payloads, one (1) payload for each target.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight capability, the UAV's payload carrying capacity and unloaded weight, making an accurate drop as close as possible to the center of the target, picking up the payload left on the hexagonal target and dropping it on the triangular target, and the time to complete the lap are parameters affecting the second mission score.
- UAVs are obliged to carry two (2) payloads within the scope of the second mission and can make two (2) different payload attempts in this mission. Teams are obliged to determine the weights of the said payloads in advance and notify the judges at the technical control (e.g., first payload attempt 800 grams and second payload attempt 1200 grams).

- A payload painted red on the outside is released onto the blue regular hexagonal target with each side measuring 2 meters, and a payload painted blue on the outside is released onto the red equilateral triangular target with each side measuring 1 meter. Since the target placements in the field aren't made according to a specific order, teams perform the payload release process on the targets regardless of the order. The teams release the payload of the appropriate color on whichever target they detect first.
- Teams performing the second mission in the International UAV Competition Rotary Wing category are deemed to have successfully completed the third mission, from which they receive bonus points as specified in the scoring, if they go to the target where they dropped their first payload after dropping two payloads, pick up the payload on the ground, and drop it on the other target.
- The distance of the payload to the target is measured based on where the payload last stops, not the place where it falls. The distance to the target is measured from the center of the target. For rotary-wing UAVs, target measurement is made up to 10 meters from the center of the target. No distance measurement is performed for payloads that fall outside these limits, and teams receive zero (0) points from this scoring criterion in the relevant mission.
- Image processing to be performed during the mission can be carried out from the air or ground without any lower altitude limitation. Teams are obliged to prove to the flight judges via the ground control station that they have detected the target via image processing method. Flight logs aren't requested from teams after the completion of flights. Even if a hit is achieved on the target without image processing, teams can't receive target hit points.
- Teams can use a single camera integrated into the auxiliary computer they use for image processing. Apart from this, using a second camera to view the area is strictly prohibited.
- Teams in Fixed Wing and Rotary Wing categories use the same runway. Rotary-wing UAVs should not detect the 4 x 4 meter and 2 x 2 meter targets positioned in the flight area for fixed-wing UAVs while performing image processing.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time

starts when the judge says "start" and is completed when the team pilot says "we are ready".

- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the second mission is ten (10) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- All teams must perform the target detection after turning from the outside of Pole 2 while performing their flights in the target detection phase in the second mission.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.

- Teams can make flight attempts with two (2) different payloads within the scope of the second mission. Teams trying different payloads must report all components and weight values they use in these flights at the technical control stage. Flights made with with payloads not declared in technical control are deemed invalid.
- Teams that perform different payload attempts aren't allowed to make any changes to the propulsion system, including but not limited to motors and propellers. Teams are only permitted to change batteries within the framework of the rules specified in Section "9.1 Technical Control".
- Teams are required to use the same motor and propeller used in the first mission while performing the second mission. UAVs are weighed before both missions. Except for mandatory situations such as accidents or crashes, no changes may be made to components affecting UAV's weight or the propulsion system. Teams declare UAV components in technical control and state their flight weights. The highest weight of the UAV during the competition is included in the calculation.
- It is mandatory to obtain the approval of the flight judge before the flight regarding which direction the take-off is performed.

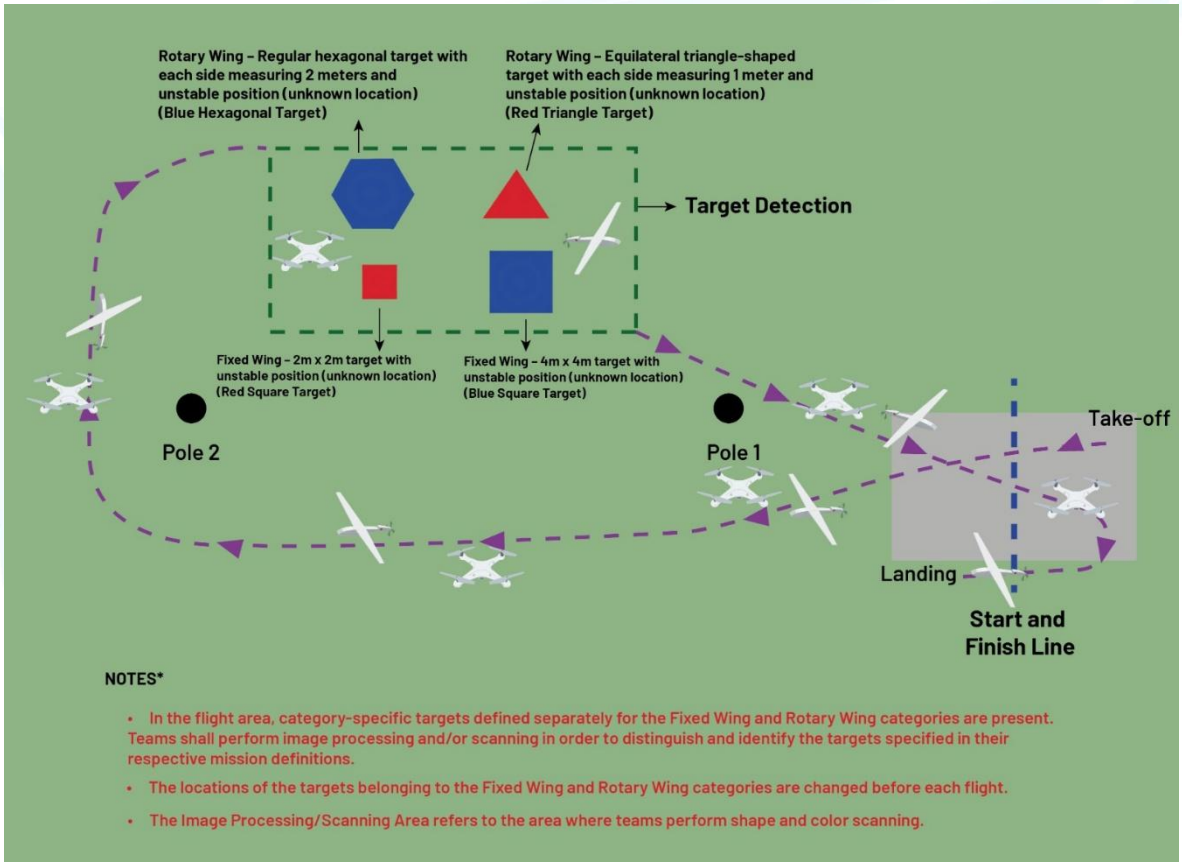

Figure 7: International UAV Competition Second Mission Flight Route

International UAV Competition Rotary Wing Category Third Mission:

- Within the scope of the second mission, payloads are released onto the triangular and hexagonal targets regardless of the order. Teams are deemed to have successfully completed third mission by receiving ten (10) points if they go to the target where they dropped their first payload and pick up the payload on the target, and by receiving ten (10) points if they pick up the payload on the target and drop it on the other target. Teams performing this mission receive bonus points as specified in the scoring.

10.2.2. Inter High School UAV Competition Rotary Wing Category Mission Descriptions

Inter High School UAV Competition Rotary Wing Category First Mission:

- In this mission, the autonomous flight capability of the rotary-wing UAV developed by the teams, the volume of the rectangular or square prism-shaped storage box, the speed of removing the UAV from the rectangular or square prism-shaped storage box and making it ready for flight, its lightness, flight speed, stable flight, and maneuverability are tested.
- The volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, autonomous flight, the unloaded weight of the UAV, and the time to complete the lap are parameters affecting the first mission score.
- Rotary-wing UAVs must perform the ∞ (figure-eight, infinite) movement by leaving the reference poles numbered 1 and 2 placed at the turning points in the flight area inside in every two (2) laps in an unloaded manner.
- UAVs fly the first mission unloaded. The first mission scoring is based on the unloaded UAV weight.
- Teams that can't perform the maneuver movement even though they meet other requirements (autonomy, fast flight, etc.) receive zero (0) points from the first mission, and teams that don't successfully complete this mission don't have the right to try the second mission.
- In the Inter High School UAV Competition, the processes of obtaining coordinates to be used in missions requiring autonomy, entering them into the system, and retrieving coordinates from the system must be carried out by team members, excluding Advisor Teachers, during the preparation stage in the competition area.

- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.
- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the first mission is five (5) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.

- UAVs take off from the take-off line. Following the take-off, the UAV proceeds directly to Pole 2 and performs a flight maneuver in the shape of infinity (∞) between two poles by turning from the outside of Pole 2. This maneuver is applied as two consecutive laps. After completing two laps, teams will pass behind Pole 2 as specified in the figure, turn from the outside of the pole, and end their missions by crossing the start/finish line to complete the track.

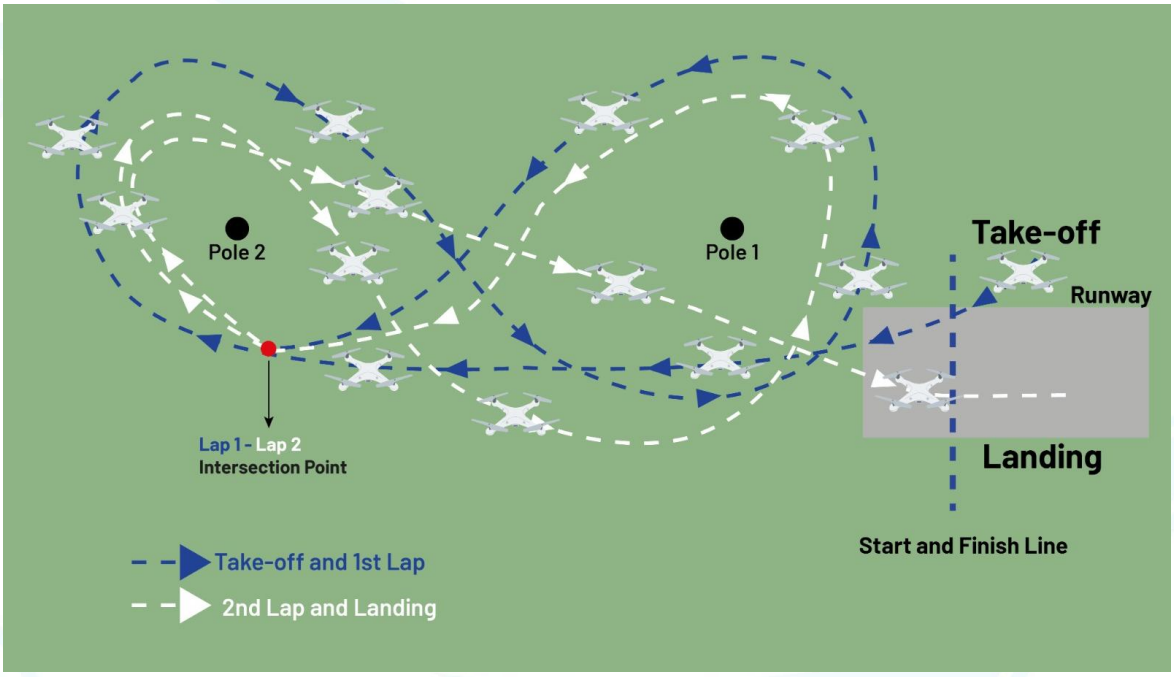

Figure 8: Inter High School UAV Competition Rotary Wing Category First Mission Flight Route

<u>Inter High School UAV Competition Rotary Wing Category Second Mission:</u>

- In this mission, teams are required to remove the rotary-wing UAV they developed from the rectangular or square prism-shaped storage box and make it ready for flight, perform autonomous flight with the flight control software, detect the locations of two (2) different targets whose locations are unknown in the flight area via image processing technique, and drop two (2) different payloads, one (1) payload for each target.
- The autonomous flight capability of the UAV, the volume of the UAV storage box, the time to remove the UAV from the storage box and make it ready for flight, the UAV's payload carrying capacity and unloaded weight, making an accurate drop as close as possible to the center of the target, and the time to complete the lap are parameters affecting the second mission score.

- UAVs are obliged to carry two (2) payloads within the scope of the second mission and can make two (2) different payload attempts in this mission. Teams are obliged to determine the weights of the said payloads in advance and notify the judges at the technical control (e.g., first payload attempt 800 grams and second payload attempt 1200 grams).
- A payload painted red on the outside is released onto the 2 x 2 meter blue target, and a payload painted blue on the outside is released onto the 1 x 1 meter red target.
- The distance of the payload to the target is measured based on where the payload last stops, not the place where it falls. The distance to the target is measured from the center of the target. For rotary-wing UAVs, target measurement is made up to 10 meters from the center of the target. No distance measurement is performed for payloads that fall outside these limits, and teams receive zero (0) points from this scoring criterion in the relevant mission.
- Image processing to be performed during the mission can be carried out from the air or ground without any lower altitude limitation. Teams are obliged to prove to the flight judges via the ground control station that they have detected the target via image processing method. Flight logs aren't requested from teams after the completion of flights. Even if a hit is achieved on the target without image processing, teams can't receive target hit points.
- Teams can use a single camera integrated into the auxiliary computer they will use for image processing. Apart from this, using a second camera to view the area is strictly prohibited.
- Teams in Fixed Wing and Rotary Wing categories use the same runway.
- Maximum time limits are defined for UAV installation, preparation for flight and mission flights in the competition.
- The upper limit for the UAV installation time is eight (8) minutes. Teams that fail to complete the installation within this time lose their flight right. The installation time starts when the judge says "start" and is completed when the team pilot says "we are ready".
- The installation process includes activities such as the assembly of UAV components, placement of payloads, and connection of power systems. No power system of the UAV may be active while the UAV is inside the storage box.

- The upper limit for the flight preparation time is eight (8) minutes. Elements such as GPS connection, arming, and similar procedures aren't included in the installation process and are considered within the scope of the flight preparation process.
- During the installation and flight preparation processes, a maximum of three (3) team members are allowed to participate in removing the UAV from the storage box and preparing it for flight. In addition to these three (3) members, one (1) additional team member may participate in the preparation of the ground control station (computer, etc.). Advisor Teachers and Advisors aren't permitted to participate in those processes.
- Following the completion of the installation and flight preparation processes, teams aren't allowed to make any additional preparations for the flight. All elements required for the flight of the UAV must be completed within the defined installation and preparation times.
- The time consisting of "duration of each additional intervention + 10 seconds" during the preparation for flight, other than powering on/off the UAV, is added to the total installation time.
- The maximum flight time for the second mission is ten (10) minutes. Teams exceeding the determined time limits are deemed unsuccessful in the relevant mission flight.
- The flight time starts when the UAV departs from the start/finish line and stops when the UAV crosses the start/finish line after performing its mission. If the UAV is damaged or crashes after crossing the start/finish line, the flight is deemed unsuccessful.
- All teams must perform the target detection after turning from the outside of Pole 2 while performing their flights in the target detection phase in the second mission.
- Choosing the landing location after crossing the finish line is left to the discretion of the teams. Teams can perform their landings in a suitable area between Pole 1 and Pole 2.
- Teams can make flight attempts with two (2) different payloads within the scope of the second mission. Teams trying different payloads must report all components and weight values they use in these flights at the technical control stage. Flights made with with payloads not declared in technical control are deemed invalid.
- Teams that perform different payload attempts aren't allowed to make any changes to the propulsion system, including but not limited to motors and propellers. Teams

are only permitted to change batteries within the framework of the rules specified in Section “9.1 Technical Control”.

- Teams are required to use the same motor and propeller used in the first mission while performing the second mission. UAVs are weighed before both missions. Except for mandatory situations such as accidents or crashes, no changes may be made to components affecting UAV’s weight or the propulsion system. Teams declare UAV components in technical control and state their flight weights. The highest weight of the UAV during the competition is included in the calculation.
- It is mandatory to obtain the approval of the flight judge before the flight regarding which direction the take-off is performed.

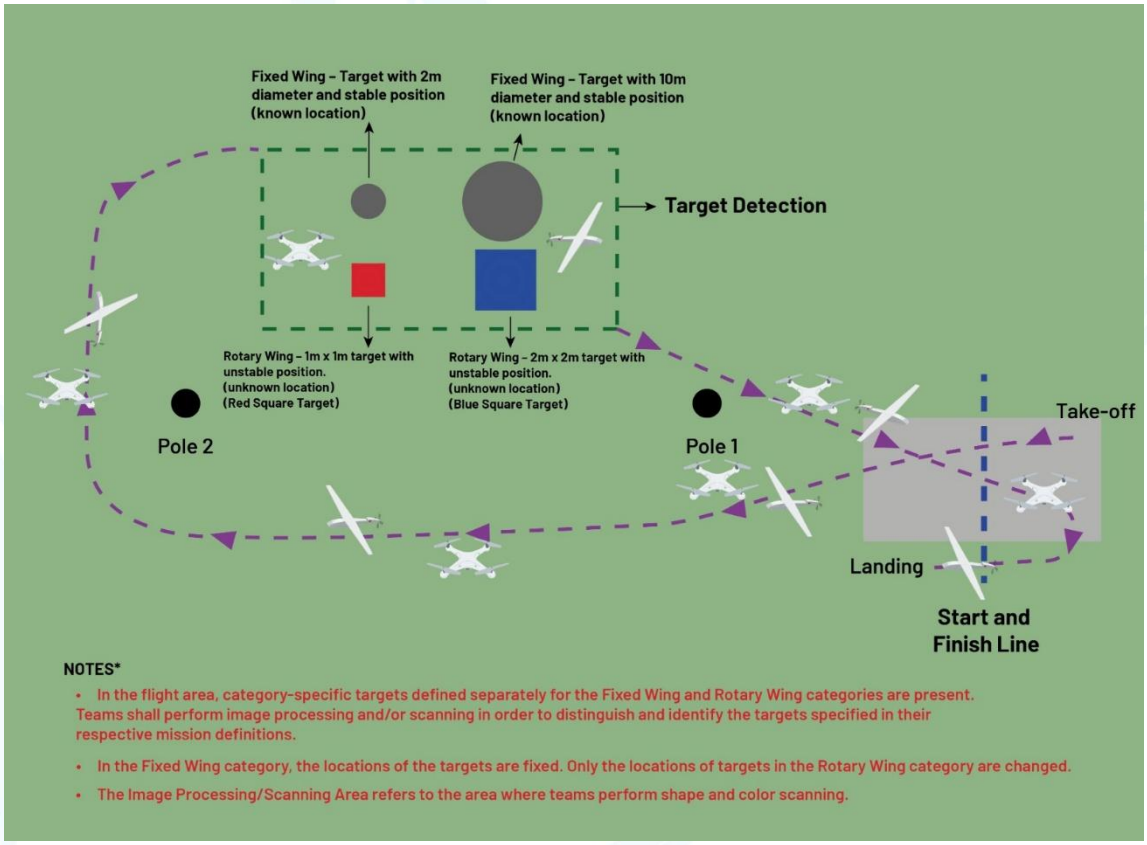

Figure 9: Inter High School UAV Competition Second Mission Flight Route

10.3. International and Inter High School UAV Competitions Free Mission Category

- In the Free Mission category of International and Inter High School UAV Competitions, no mission definition is provided. The mission is determined entirely

by the teams. The exact same scenario in the Preparation and Mission Video is performed in the competition area.

- The flight time should not exceed 15 minutes. Teams must give a short briefing about their flights and missions, not exceeding 5 minutes, to the flight judges before the flight.
- The mission to be performed in the area is compared with the mission specified in the Project Presentation Report and Preparation and Mission Video. A different mission isn't accepted.
- Flights can be performed manually or autonomously. However, autonomously performed missions receive higher points.
- Determining and achieving a different and challenging mission is essential in competitions. Missions performed before or similar to each other affect the success of the teams.
- Teams determine a mission flight aiming to collect the maximum points by considering the headings and scores specified in the evaluation criteria. However, some teams may face the application rejection or the request for alteration of mission definition (e.g., using a paint gun to shoot instead of a real weapon mechanism, or imposing some restrictions due to safety for a team wishing to perform high-speed acrobatics) taking into account
  - safety,
  - flight area constraints,
  - sonstraints specified in the competition rules.
- Teams that can't fully perform their mission in the competition area can't qualify to receive a Ranking Award.

11. SCORING

Flights performed in the competition area are evaluated and scored by judges who are experts in their fields according to the formulas or evaluation criteria specified below.

Any statement or actual action that influences or misleads the judges and officials assisting the judges is prohibited.

In the competition, in addition to technical gains, the behaviors of the team members and their attitudes towards TÜBİTAK and TEKNOFEST officials, judges, officials assisting the judges, THK pilots, and other team members in the area are also included in the evaluation and reflected in the scoring.

In this context, competitions require the observance of social values such as interdisciplinary work, sustainability, process discipline, honesty, openness, helpfulness, and respect for the success and rights of others, in addition to technical gains. Teams acting contrary to these values, abusing preparation and mission flexibility, engaging in statements or actions intended to influence or mislead judges and officials assisting them are considered unsportsmanlike conduct. In case of detection of such and similar situations, a penalty score is applied over the total score of the teams.

Team awareness refers to the team members acting in a safe, disciplined, and coordinated manner towards a common goal throughout the competition. In this context, clarity of duties and responsibilities within the team, effective communication, management of time and resources in accordance with mission and safety priorities, and full compliance with competition rules and safety procedures are taken as a basis. Attitudes that contradict these principles and unsportsmanlike behaviors are considered detrimental to team awareness.

In case of detection of such and similar situations,

- In the Free Mission category, in case of detection of behaviors contrary to the team awareness criterion and/or sportsmanship, a penalty score deduction between zero (0) and five (5) points is applied from the team awareness criterion.
- In Fixed Wing and Rotary Wing categories, in case of detection of unsportsmanlike behaviors, a penalty score deduction between zero (0) and five (5) points is applied over the total score of the teams.

If teams continue their negative attitudes despite the warnings, they may be eliminated from the competition and/or removed from the competition area by the decision of the AAC.

Teams participating in the competition with an aircraft that doesn't comply with the specification, running motors or performing flight attempts in unauthorized areas, performing hardware or software changes without the approval of the AAC and/or flight judge throughout the competition process, directly or indirectly sabotaging the flight or systems of other teams, and also teams detected to have made false or misleading declarations in team information, licenses, permits, or official documents are eliminated from the competition.

11.1. International UAV Competition Fixed Wing and Rotary Wing Categories

11.1.1. General Scoring Table

Table 2: International UAV Competition Fixed Wing and Rotary Wing Categories General Scoring Table

	Preparation and Mission Video Score	Mission 1 Score	Mission 2 Score	Total Score	Penalty Score 1	Penalty Score 2
Fixed Wing Category	10	30	70	110	0-5	0-10
Rotary Wing Category	10	30	80	120	0-5	0-10

11.1.2. Mission Scoring Table

Table 3: International UAV Competition Fixed Wing and Rotary Wing Categories Mission Scoring Table

Mission	Formula
Mission 1 (Fixed & Rotary Wing)	$M_{S1} = c.\ [6.\ \frac{Vol_{min}}{Vol_{tk}} + 6.\ \frac{t_{min,prep}}{t_{tk,prep}} + 6.\ \frac{W_{0,min}}{W_{0,tk}} + 12.\ \frac{t_{min,flight}}{t_{tk,flight}}]$ - Penalty Score
Mission 2 (Fixed Wing)	$M_{S2} = c.\ [4.\ \frac{Vol_{min}}{Vol_{tk}} + 4.\ \frac{t_{min,prep}}{t_{tk,prep}} + 4.\ \frac{W_{0,min}}{W_{0,tk}} + 20.\ \frac{W_{payload,tk}}{W_{payload,max}} + 10.\ \frac{d_{4x4,min}}{d_{4x4,tk}} + 20.\ \frac{d_{2x2m,min}}{d_{2x2m,tk}} + 8.\ \frac{t_{min,flight}}{t_{tk,flight}}]$ - Penalty Score
Mission 2 (Rotary Wing)	$M_{S2} = c.\ [4.\ \frac{Vol_{min}}{Vol_{tk}} + 4.\ \frac{t_{min,prep}}{t_{tk,prep}} + 4.\ \frac{W_{0,min}}{W_{0,tk}} + 20.\ \frac{W_{payload,tk}}{W_{payload,max}} + 8.\ \frac{d_{hex,min}}{d_{hex,tk}} + 12.\ \frac{d_{tri,min}}{d_{tri,tk}} + 10.\ (AY_{hex,payload} + BY_{tri,payload}) + 8.\ \frac{t_{min,flight}}{t_{tk,flight}}]$ - Penalty Score

11.1.3. Scoring Parameters

MS1: Mission 1 Score

MS2: Mission 2 Score

c: Autonomous flight coefficient value is evaluated as follows:

- 0.7 in case of manual take-off and/or manual landing, autonomous flight in the International UAV Competition,
- 1,0 in case of autonomous take-off, autonomous flight, and autonomous landing in the International UAV Competition.

Vol_{min}: The lowest UAV storage box volume among all teams

Vol_{tk}: The volume of the team's UAV storage box

$t_{min,prep}$: The lowest UAV installation time for preparation for flight among all teams

$t_{tk,prep}$: The team's UAV installation time for preparation for the relevant flight

$t_{min,flight}$: The lowest flight time among all teams successful in the relevant mission

$t_{tk,flight}$: The team's time to perform the relevant mission

$W_{0,min}$: The lowest unloaded weight among all teams

$W_{0,tk}$: The team's unloaded weight

$W_{yük,tk}$: The weight of the payload to be carried by the team

$W_{yük,max}$: The highest total payload weight among all teams

$d_{4x4,min}$: The minimum distance of the dropped payload to the center of the 4 x 4 meter target among all teams (Fixed wing)

$d_{4x4,tk}$: The distance of the payload dropped by the team to the center of the 4 x 4 meter target (Fixed wing)

$d_{2x2,min}$: The minimum distance of the dropped payload to the center of the 2 x 2 meter target among all teams (Fixed wing)

$d_{2x2,tk}$: The distance of the payload dropped by the team to the center of the 2 x 2 meter target (Fixed wing)

$d_{hex,min}$: The minimum distance of the dropped payload to the center of the hexagonal target among all teams (Rotary wing)

$d_{hex,tk}$: The distance of the payload dropped by the team to the center of the hexagonal target among all teams (Rotary wing)

$d_{tri,min}$: The minimum distance of the dropped payload to the center of the triangular target among all teams (Rotary wing)

$d_{tri,tk}$: The distance of the payload dropped by the team to the center of the triangular target (Rotary wing)

$AY_{hex,payload}$: Payload picked up from the hexagonal target (failed to pick up: 0, picked up: 1)

$BY_{tri,payload}$: Payload dropped onto the triangular target (miss: 0, hit: 1)

Penalty Score 1: Penalty score to be applied for teams displaying unsportsmanlike behaviors and for materials detected not to be in the storage box and added to the UAV later.

Penalty Score 2: Penalty score to be applied for teams taking the pole from the inside once / not turning from the outside of the pole once.

11.2. Inter High School UAV Competition Fixed Wing and Rotary Wing Categories

11.2.1. General Scoring Table

Table 4: Inter High School UAV Competition Fixed Wing and Rotary Wing Categories General Scoring Table

	Preparation and Mission Video Score	Mission 1 Score	Mission 2 Score	Total Score	Penalty Score 1	Penalty Score 2
Fixed Wing and Rotary Wing Categories	10	30	70	110	0-5	0-10

11.2.2. Mission Scoring Table

Table 5: Inter High School UAV Competition Fixed Wing and Rotary Wing Categories Mission Scoring Table

Mission	Formula
Mission 1 (Fixed & Rotary Wing)	M_{S1} = c. [6. $\frac{Vol_{min}}{Vol_{tk}}$ + 6. $\frac{t_{min,prep}}{t_{tk,prep}}$+ 6. $\frac{W_{0,min}}{W_{0,tk}}$+12. $\frac{t_{min,flight}}{t_{tk,flight}}$] - Penalty Score
Mission 2 (Fixed Wing)	M_{S2} = c. [4. $\frac{Vol_{min}}{Vol_{tk}}$ + 4. $\frac{t_{min,prep}}{t_{tk,prep}}$+ 4. $\frac{W_{0,min}}{W_{0,tk}}$+20. $\frac{W_{payload,tk}}{W_{payload,max}}$+10. $\frac{d_{10m,min}}{d_{10m,tk}}$+ 20. $\frac{d_{2m,min}}{d_{2m,tk}}$ + 8. $\frac{t_{min,flight}}{t_{tk,flight}}$] - Penalty Score
Mission 2 (Rotary Wing)	M_{S2} = c. [4. $\frac{Vol_{min}}{Vol_{tk}}$ + 4. $\frac{t_{min,prep}}{t_{tk,prep}}$+ 4. $\frac{W_{0,min}}{W_{0,tk}}$+20. $\frac{W_{payload,tk}}{W_{payload,max}}$+10. $\frac{d_{2x2m,min}}{d_{2x2m,tk}}$+ 20. $\frac{d_{1x1m,min}}{d_{1x1m,tk}}$+8. $\frac{t_{min,flight}}{t_{tk,flight}}$] - Penalty Score

11.2.3. Scoring Parameters

MS1: Mission 1 Score

MS2: Mission 2 Score

c: Autonomous flight coefficient value is evaluated as follows:

- 0.5 in case of manual take-off, manual flight and manual landing in the Inter High School UAV Competition,
- 0.7 in case of manual take-off and/or manual landing, autonomous flight in the Inter High School UAV Competition,
- 1,0 in case of autonomous take-off, autonomous flight and autonomous landing in the Inter High School UAV Competition.

Vol_{min}: The lowest UAV storage box volume among all teams

Vol_{tk}: The volume of the team's UAV storage box

$t_{min,prep}$: The lowest UAV installation time for preparation for flight among all teams

$t_{tk,prep}$: The team's UAV installation time for preparation for the relevant flight

$t_{min,flight}$: The lowest flight time among all teams successful in the relevant mission

$t_{tk,\ flight}$: The team's time to perform the relevant mission

$W_{0,min}$: The lowest unloaded weight among all teams

$W_{0,tk}$: The team's unloaded weight

$W_{payload,tk}$: The weight of the payload to be carried by the team

$W_{payload,max}$: The highest total payload weight among all teams

$d_{10m,min}$: The minimum distance of the dropped payload to the center of the 10-meter diameter target among all teams (Fixed wing)

$d_{10m,tk}$: The distance of the payload dropped by the team to the center of the 10-meter diameter target (Fixed wing)

$d_{2m,min}$: The minimum distance of the dropped payload to the center of the 2-meter diameter target among all teams (Fixed wing)

$d_{2m,tk}$: The distance of the payload dropped by the team to the center of the 2-meter diameter target (Fixed wing)

$d_{2x2m,min}$: The minimum distance of the dropped payload to the center of the 2 x 2 meter target among all teams (Rotary wing)

$d_{2x2m,tk}$: The distance of the payload dropped by the team to the center of the 2 x 2 meter target (Rotary wing)

$d_{1x1,min}$: The minimum distance of the dropped payload to the center of the 1 x 1 meter target among all teams (Rotary wing)

$d_{1x1,tk}$: The distance of the payload dropped by the team to the center of the 1 x 1 meter target (Rotary wing)

Penalty Score 1: Penalty score to be applied for teams displaying unsportsmanlike behaviors and for materials detected not to be in the storage box and added to the UAV later.

Penalty Score 2: Penalty score to be applied for teams taking the pole from the inside once / not turning from the outside of the pole once.

11.3. International and Inter High School UAV Competitions Free Mission Category

11.3.1. Evaluation Criteria and Evaluation Method

Table 6: International and Inter High School UAV Competitions Free Mission Category Scoring Table

	Main Criterion	Sub-Criteria	Explanation	Max Score	
				Inter High School UAV Competition	International UAV Competition
1	**Autonomy**	Mission Planning Autonomy	Are route, target selection, decision algorithms autonomous?	0–3	0–3
		Environmental Perception and Adaptation	Changing route/mission according to camera and sensor data	0–2	0–2
		Continuous Autonomous Mission Completion	Continuity from the beginning to the end of the mission	0–10	0–10
2	**Stability**	Take-off Stability	Uncontrolled take-off and climb	0–3	0–3
		Cruise Stability	Heading deviation, yaw, pitch, flutter	0–4	0–4
		Landing Stability	Uncontrolled descent and hard landing	0–3	0–3
3	**Capability**	Mission Type Difficulty Coefficient	Difficulty coefficient	0–10	0–10
		Mission Success Degree	Success rate of mission objectives	0–5	0–5
		Mission Performance Efficiency	Time / accuracy / energy consumption	0–5	0–5
		Adaptation to Condition Change	Wind, signal, obstacles, etc.	0–5	0–5
4	**Mission Effectiveness & Demo Quality**	Mission Story Consistency	Problem definition – solution relationship	0–3	0–3
		Demonstration Clarity	Judges understanding the flow	0–3	0–3
		User Impact / Applicability	Transfer to practical life	0–4	0–4
5	**Domestic Product Usage**[1]	Software Domestic Contribution Rate	Domestic production contribution of the software	0–5	0–5
		Mechanical/Electronic Domestic Contribution	Domestic production contribution of the design	0–5	0–5
6	**Innovation &**	Original Mission Scenario	New mission approach	0–5	0–5

	Originality[1]	Technological Innovation	New sensor, algorithm, mechanical design	0–5	0–5
		Scientific/Industrial Contribution	Publication, patent, R&D	0–5	0–5
7	**Economy & Simplicity**	Use of Unnecessary Hardware	Less equipment, high efficiency	0–5	0–5
		Cost/Performance Ratio	Estimated cost – efficiency relationship	0–5	0–5
8	**Team Awareness**	Teamwork/Crew Resource Management (CRM)	Ability to act together, work sharing, leadership and communication skills	0-2	0-2
		Time Management	Teams' ability to use time efficiently and effectively	0-1	0-1
		Compliance with Rules	Compliance with all rules and regulations, discipline shown during the competition, harmony/communication with other teams and officials	0-2	0-2
			TOTAL SCORE:	**100**	**100**

[1] *The responsibility of all kinds of information, documentation, and convincing the judges regarding "Innovation & Originality" and "Domestic Product Usage" belongs to the competitors. Especially original works related to software must be documented. In order to receive points from the "Innovation & Originality" section, the said elements must not have been declared in another competition.*

Autonomy:

It is expected that take-offs, flights, and landings are autonomous while performing the mission. Flights controlled entirely by the controller can't receive points from the "Continuous Autonomous Mission Completion" sub-criterion.

"Mission Planning Autonomy" sub-criterion refers to the level of route determination, target selection, making alternative choices, and running decision algorithms on its own during the mission. If the system can plan without human command, it receives a high score.

"Environmental Perception and Adaptation" sub-criterion measures the ability of the UAV to perceive its surroundings with sensors and make route or mission changes according to this information. Being able to adapt to changes such as wind, obstacles, and target loss is important.

"Continuous Autonomous Mission Completion" sub-criterion evaluates the vehicle's ability to complete the mission continuously, stably, and without human intervention from start to finish. If there is deviation and pause, or manual intervention is required during the mission, the score decreases.

Stability:

This criterion evaluates the level of keeping the flight dynamics of the UAV under control. The aim is to measure safe, trouble-free movement performance without loss of control. In the "Take-off Stability" sub-criterion, it is evaluated whether the UAV experiences problems such as uncontrolled banking, sudden climb, or unstable take-off during the take-off process.

Within the scope of the "Cruise Stability" sub-criterion, it is checked whether there are situations such as directional instability, unnecessary wobbling, excessive oscillation, or forward-backward control disorders during the flight. The more stable and smooth the UAV flies, the higher the team's score is.

"Landing Stability" sub-criterion indicates whether the vehicle experiences hard landing, side diving, loss of control, or unstable descent during landing.

Example 1: Rotary-wing UAVs must maintain the hover position for at least 10 seconds and fixed-wing UAVs must perform a straight and stable flight for at least 10 seconds without yawing, deviating, or pitching.

Capability:

Regardless of the class of the UAV and the mission it performs, the mission performed has a difficulty degree and there is an ability to achieve this mission. With this criterion, how effectively, quickly, accurately, and in accordance with the rules the UAV performs the mission is evaluated. Since missions vary greatly, it is normalized with the "Mission Type Difficulty Coefficient".

"Mission Type Difficulty Coefficient" sub-criterion is a multiplier determined by the judges in advance, expressing the general difficulty level of the mission. More complex missions are evaluated with a higher coefficient.

"Mission Success Degree" sub-criterion shows what percentage of the targets and steps given in the mission have been successfully completed. If most of the targets are achieved completely, the score increases.

"Mission Performance Efficiency" sub-criterion is a measure evaluating how appropriately and effectively the time spent, the number of errors made, and the energy or other resources are used during the execution of a mission. This concept expresses completing the mission both correctly and with minimum effort.

With the "Adaptation to Condition Change" sub-criterion, how the vehicle reacts to situations such as wind change, signal power loss, sudden obstacles, GPS error is evaluated. Adaptable systems receive high scores.

Mission Effectiveness & Demo Quality:

How understandable, traceable, and impressive the presentation and mission demonstration are by the judges is measured.

With the "Mission Story Consistency" sub-criterion, it is evaluated whether the team conveys the questions "what is the problem, what is the solution, why this way" in a logical and consistent manner. At the same time, the compatibility of the mission presented in the Project Presentation Report with the mission presented in the competition area is checked.

Within the scope of the "Demonstration Clarity" sub-criterion, it is important for the judges to easily understand the mission flow, for the narration to be clear, and for the UAV's actions to be seen clearly.

In the "User Impact and Applicability" sub-criterion, the applicability of the mission to real life, the benefit it creates for the user, and the potential for practical use are measured.

Domestic Product Usage:

The domestic contribution rate of the team on both software and hardware is measured. Within the scope of the "Mechanical/Electronic Domestic Contribution" sub-criterion; it is evaluated how much of the UAV's fuselage and structural components (wing, fuselage, landing gear, etc.), electronic cards (power, communication, sensor interfaces), integration and assembly solutions of sensors, and other physical subsystems are developed through an R&D process carried out in the team's country and/or produced by domestic manufacturers. Ready-made components procured only from domestic companies but whose design and development process isn't carried out in the country aren't accepted as domestic contributions in this context.

Domestic Product Usage Example: The development/production of a hardware, software, or auxiliary equipment by domestic manufacturers in the team's country is an indicator of domestic production. In this section, hardware and software components developed or produced in the country and procured from domestic manufacturers are evaluated. Buying only from domestic companies isn't sufficient alone for the relevant element to be evaluated within the scope of the "Domestic Product Usage" criterion. For a hardware or software component to be included in the "Domestic Product Usage" criteria, the product in question must be personally developed or produced by the domestic company. Accordingly, products such as **filament, screw, foam adhesive,**

foreign-origin autopilot systems procured from domestic companies but not meeting the above criteria aren't evaluated within the scope of "Domestic Product Usage".

Innovation & Originality:

This criterion measures the degree of technical and conceptual innovation exhibited by the team. The UAV is expected to contain innovative sensors, new algorithms, special mechanical designs, or original software solutions as much as possible, or the team is expected to improve the existing system by adding new features.

"Originality" is the use of a new system or concept that did not exist before. Within the scope of the "Original Mission Scenario" sub-criterion, creative mission fictions that solve the problem differently and have not been seen in previous competitions receive high scores.

Within the scope of the "Technological Innovation" sub-criterion, innovative sensors, new algorithms, special mechanical designs, or original software solutions are evaluated.

For example:

- A different energy source (Using new battery types different from batteries used today)
- An original software or controller (Using the team's own software or controller instead of ready-made software or controllers for UAVs)
- A different field (Aiming to use aircraft in a new field other than their current usage areas)
- An original design (Aircraft having different designs other than those in the literature and being structurally original)
- Use of original auxiliary equipment (Use of equipment such as aircraft tracking antenna, launch system, take-off system, etc.)

Within the scope of the "Scientific/Industrial Contribution" sub-criterion, it is evaluated whether original engineering solutions are presented in each of the design, production, control, navigation, detection, and mission execution processes of the developed UAV. It is considered which technical and methodological aspects of the project distinguish it from approaches in the literature for all missions performed by the UAV, whether it includes an experimental or analytical R&D process, and whether this process is concretized with flight tests, mission success rates, performance improvements, and verified data. Also, the potential of its outputs to produce scientific publications, make

patent or utility model applications, and provide applicable, measurable technical or economic contributions in defense, aviation, or civil industries is taken into account.

Economy & Simplicity:

This criterion measures how effectively the mission can be done with minimum cost and equipment without complicating it unnecessarily.

Using additional hardware that isn't needed for the mission or doesn't provide efficiency causes a score decrease in the "Use of Unnecessary Hardware" sub-criterion. The "less but functional equipment" approach is encouraged.

In the "Cost/Performance Ratio" sub-criterion, the balance between the system cost and the obtained mission performance is evaluated. Teams providing high success with affordable cost are advantageous.

Team Awareness:

Team awareness is an important phenomenon aiming not only technical gains but also social and ethical gains in competitions.
In aviation competitions, beyond individual competencies, the capacity of team members to act in a safe, disciplined, and coordinated manner towards a common goal comes to the fore.
This process is based on values such as interdisciplinary work, ensuring sustainability through the joint work of lower and upper classes, honesty, openness, and transparency.
Team awareness requires the common goal to be understood in the same way by the whole team, and roles to be clearly determined and internalized.
Effective communication, mutual trust, awareness of responsibility, and competence to act together in moments of crisis are the basic elements of this consciousness. In a field where error tolerance is extremely low such as aviation, preserving process discipline and managing the process safely is of critical importance.
In this context, team awareness covers not only technical success but also managing the process in a sustainable and safe manner. The team awareness criterion consists of 3 sub-criteria as "Teamwork/Crew Resource Management (CRM)", "Time Management", and "Compliance with Rules".
"Teamwork/Crew Resource Management (CRM)" sub-criterion is a systematic approach aiming to use all available resources such as human, equipment, time, information, and environmental conditions in the most effective and safe way to increase safety and

performance in aviation. CRM focuses on behavioral competencies such as effective communication, situational awareness, decision making, leadership and followership, workload and stress management. In this aspect, CRM is a management and education process that goes beyond individual technical competencies, improves the capacity of team members to prevent and manage errors, and centers on safety culture.

"Time Management" sub-criterion means managing time in accordance with safety, performance, and mission priorities during preparation, implementation, and evaluation stages throughout the competition. In such competitions, time is a critical operational factor directly affecting decision-making and coordination processes beyond being a technical constraint. During the competition, time management is possible by adhering to the predetermined mission flow, clarity of role distribution, and short, standardized communication. The basic principle isn't to compromise safety under time pressure and to make critical decisions in accordance with procedures, not in a hurry. Time management is evaluated by analyzing mission duration, delays, and recovery speeds. In this context, effective time management in aviation competitions reflects the team's competence to do the right job, in the right order, and at the right time, rather than individual speed.

"Compliance with Rules" sub-criterion is the basic condition for the competition to be conducted in a fair, safe, and standard manner and reflects team discipline and professionalism level independent of technical success. Compliance with rules covers reading and correctly interpreting competition instructions completely, strictly adhering to safety procedures, not exceeding technical and operational limits, complying with time and mission constraints, and adopting ethical behaviors. In this context, compliance with rules is evaluated not only to avoid punishment but as an indicator of internalizing the understanding of safety and standardization, which are the basic elements of aviation culture.

In the Free Mission category, evaluation is made over eight (8) criteria, and each criterion is evaluated by at least three (3) judges. All judges of Free Mission category share their opinions to the judges scoring for "Mission Effectiveness & Demo Quality", "Domestic Product Usage", "Innovation & Originality", "Economy & Simplicity", and "Team Awareness" criteria. Said opinions are evaluated by the scoring judges. The sum of the average scores obtained constitutes the total score of that team.

Table 7: Free Mission Category Scoring Example

Scoring Example for a Team (J: Judge)																					
	J 1	J 2	J 3	J 4	J 5	J 6	J 7	J 8	J 9	J 10	J 11	J 12	J 13	J 14	J 15	J 16	J 17	J 18	J 19	J 20	Avg.
Criterion 1	15	9	12																		12
Criterion 2			8	7	6																7
Criterion 3						17	22	21													20
Criterion 4									7	8	9										8
Criterion 5												2	7	3							4
Criterion 6														12	8	10					10
Criterion 7																8	5	5			6
Criterion 8																		2	2	5	3
																				TOTAL:	70

In the competition area, excluding reasons caused by adverse weather conditions, teams whose UAV can't fly can't receive points from the evaluation criteria. In order to receive points in the competition area, the UAV must fly successfully. By the expression of the UAV flying successfully, it is meant that the UAV holds on in the air for a while (becomes airborne). It is concluded that UAVs that can't hold on in the air or crash before holding on in the air can't fly.

While scoring a institution or a team at least one of whose members belongs to that institution, a judge who is an employee of that institution temporarily leaves their duty to another judge.

When the first and second rounds are completed, the ranking is made over the total scores and the scores are announced. When the third round is completed, the higher total scores of the teams that used the second and third flight rights are taken as a basis, and the final results are announced by making a ranking.

12. AWARDS

- In the event that the finalist teams participating in the competition without receiving any financial support from TUBITAK and do not receive any Ranking Award, they compete for the Performance Awards calculated in proportion to their flight scores in the competition area. A finalist team that participates in the competition without receiving financial support and receives a Ranking Award can't receive a Performance Award.
- Teams that successfully complete Mission 1 in the Fixed Wing and Rotary Wing categories are able to move on to Mission 2. Teams that fail to successfully complete Mission 2 can't be eligible to receive a Ranking Award.
- Teams that can't fully perform their mission in the Free Mission category can't qualify to receive a Ranking Award.
- The awards stated in the table below show the total amount to be given to the teams that are entitled to receive awards. There is no individual awarding. First, second and third place awards are divided equally according to the total number of team members (excluding the Advisor and Advisor Teacher) and are paid in the competition area. Advisors and Advisor Teachers whose teams are eligible to receive ranking awards can't benefit from the first, second and third place award amounts below. The awards to be given to the Advisors and Advisor Teachers are also stated in the table below.

Table 8: Awards

Name of Award	Definition	International UAV Competition (for 3 categories)	International UAV Competition Advisor Awards	Inter High School UAV Competition (for 3 categories)	Inter High School UAV Competition Advisor Teacher Awards
Ranking Awards	First Place Award	250,000 TRY	20,000 TRY	200,000 TRY	20,000 TRY
	Second Place Award	200,000 TRY	15,000 TRY	150,000 TRY	15,000 TRY
	Third Place Award	175,000 TRY	12,000 TRY	120,000 TRY	12,000 TRY
Performance Awards	55 points and above	Point x 750 TRY	-	Puan x 550 TRY	-

12.1. TUSAŞ Special Award

- TUSAŞ Special Awards are announced later.

12.2. Payment of Awards

- Ranking Awards are shared equally among all system-registered members of the team, excluding the Advisor and Advisor Teacher.
- Advisors and Advisor Teachers can't receive a share from Ranking Awards. Advisors and Advisor Teachers of teams receiving a Rank Award receive the Advisor/Advisor Teacher Awards specified in the award table.
- Performance Awards are shared equally among all system-registered members of the team, including the Advisor and Advisor Teacher.
- Financial Project Support (if any), Ranking Awards (if any) and Performance Awards (if any) of the finalist teams participating from abroad (including Azerbaijan and TRNC) are paid to the Team Responsible in cash in the competition area or TEKNOFEST area.
- In case Financial Project Support, Ranking Awards and Performance Awards can't be paid in cash in the competition/TEKNOFEST area to teams participating from abroad, teams are obliged to provide the TÜBİTAK official with their bank accounts open to international Euro or Dollar transfers. No support and award payment shall be made in case of incorrect, incomplete, or mismatched bank information.

13. USE OF SUPPORT AND RETURNS

TÜBİTAK, conferring with the Advisory and Assessment Committee (AAC), may request a full refund of financial project support in the following cases:

- The team not entering the technical control or not performing the flights in the competition area.
- Failure to participate in the award ceremony in case of receiving an award.
- Expulsion of the team from the competition by TÜBİTAK, by taking the opinion of the AAC, due to ethical violation, technical inadequacy, or unsportsmanlike behaviors during the competition process.
- Submission of unrealistic, misleading information and documents.

14. OTHER PROVISION

14.1. Administrative Provisions

- Throughout the competitions, all teams are required to follow the announcements on the www.teknofest.org page and their e-mails registered in the application system.
- Changes can be made in the Specification if deemed necessary by TÜBİTAK. The most recently announced current Specification is valid in the competitions.
- In case of disagreement in technical rules, the decisions of the AAC are valid. In cases where there is no provision, the decisions of the Departmant are applied.
- Problems that can be solved if precautions are taken in advance, such as orders getting stuck in customs, and delays in cargo, can't be presented as excuses for not coming to the competition.
- There must be no relationship accepted as a first-degree kinship or conflict of interest between the students and the Advisor Teacher or Advisor. In case of detection, the team is disqualified from the competition.
- The accuracy of the information declared during the application can be checked by TÜBİTAK via online systems. In case of misleading and untrue declarations, the responsibility belongs to the owners of the declaration.
- In accordance with the subparagraph (i) of the first paragraph of the Article 3 of the Regulation on Programs to be carried out by the TÜBİTAK Directorate for Science and Society, there may be limitations on the award amounts to be received by individuals.
- The team may be expelled from the competition or may be banned from the competition for at least one (1) year by TÜBİTAK as a result of the evaluation of Advisory and Assessment Committee in the event that one or more of the team members applying for the competition behave aggressively or rudely towards TÜBİTAK officials, TEKNOFEST officials, AAC members, the judges, other team members or competition officials in writing, verbally or physically before, during or after the competition against including face-to-face, written or online environments.

14.2. Warnings Regarding Test Flights Before Competition Week

Teams must pay attention to the following rules for test flights during the preparation process:

- It is mandatory to have a UAV pilot among the team members.

For flights to be carried out in free (green) zones at times not prohibited by the highest local administrative authority of the locality,

- Flights can be performed only in cases where meteorological conditions are met by seeing, between sunrise and sunset hours, and in weather open to visibility of at least two (2) km.
- The UAV must be within the pilot's line of sight, not exceeding 500 meters horizontally.
- It should not go above 400 feet (120 meters) from the ground (AGL).
- Flights must be carried out at least 50 meters away from people and structures.
- Test flights are carried out by persons who have pilot authorization suitable for the class of the UAV.
- Civil authorities may declare flight bans in some regions at various periods for safety reasons. Before the test flight, it should be checked whether there is a flight ban imposed on the flight zone. According to the legal legislation, UAV owners are responsible for damages caused by UAVs to third parties.

It is the responsibility of the teams to comply with all rules determined by the civil administrative authorities in test flights.

14.3. Security Measures

The security measures specified below are mandatory to be followed in the context of both human health and human rights violations. In case of violation of these measures, the AAC has the right to impose all kinds of sanctions in line with the opinion and approval of TÜBİTAK.

- Testing vehicles that could not perform flight tests before the competition week for the first time during the competition,
- Coming to the area without performing pre-tests if significant changes have been made on the vehicle although test trials were made before,
- Attaching propellers to UAVs and giving motion in closed environments (tents etc.) or turning them at high speed and endangering the safety of others on competition days,
- While a team is performing a mission flight in the area, another team performing a test flight in the nearby region and creating a possibility or suspicion of signal jamming,
- Intervening manually in a vehicle that has fallen to the ground while its propellers continue to turn unless it is very urgent

- Failure of the pilot or responsible member to make the necessary intervention on time although the UAV goes out of the border when it takes off for the mission flight and is seen as dangerous by the judge and a landing order is given,
- Vehicles that have crashed going out for a second flight without passing through technical control again.

15. RELEVANT LEGISLATION

- Regulation on Events to be carried out by the TÜBİTAK Directorate for Science and Society,
- UAV Competitions Specification,
- In cases where there is no provision in the relevant legislation, the decisions of the Department and AAC are applied.

16. CONTACT

- Application, report and video upload: www.t3kys.com/en/
- Announcements and information: www.teknofest.org
- For your questions regarding the International UAV Competition: uavturkey@tubitak.gov.tr and https://groups.google.com/u/0/g/uluslararasi-iha-yarismasi?hl=tr
- For your questions regarding the Inter High School UAV Competition: ihaliseler@tubitak.gov.tr and https://groups.google.com/u/0/g/-liseler-arasi-iha-yarismasi?hl=tr

9TH YEAR

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